Scientific Analysis of Moral Judgments using Tobit and Cluster-Aware Non-Parametric Robustness Models

Automated Research Pipeline

marzo 28, 2026

1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from the Bucaramanga Campus. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator's decision resulted in varying degrees of payoff for themselves, their own group, and a victim group. Under Option 2: explicit case-configuration modeling , each judgment is treated as relational and indexed by an explicit victim x negotiator scenario configuration rather than by isolated ingroup/outgroup attributes.

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{ij}	Observed moral judgment of scenario j by participant i.
y^*_{ij}	Latent moral preference score (unbounded).
β_1	Regression coefficient representing the marginal effect of the predictor.
$\beta_{0.5}$	Median-regression coefficient from the non-parametric censored robustness model.
$Q_{0.5}(y^*_{ij} \mid \mathbf{x}_{ij})$	Conditional median of the latent bounded outcome given the predictors.
IRI_i	Empathy score (Average composite of the Interpersonal Reactivity Index).
$\mathrm{CaseConfig}_{ij}$	Explicit victim x negotiator case-configuration factor capturing relational scenarios such as `Hum_x_Hum`, `Hum_x_Ing`, `Hum_x_Control`, `Ing_x_Hum`, `Ing_x_Ing`, `Ing_x_Control` .
Role_{ij}	Scenario-role conditioning indicator distinguishing Observer from Victim judgments.
$\mathrm{Decision}_{ij}$	Decision-context conditioning indicator distinguishing Accept from Reject judgments.
ICC	Intraclass Correlation: ratio of between-cluster variance to total variance.

Option 2: explicit case-configuration modeling frames each judgment as a relational victim x negotiator case. The paired-group structure generates interpretable configurations such as Hum_x_Hum, Hum_x_Ing, Hum_x_Control, Ing_x_Hum, Ing_x_Ing, and Ing_x_Control, with role (Observer/Victim) and decision context (Accept/Reject) available as further conditioning dimensions.

3 Hypotheses to Test

- **H1 (Empathy):** Higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on explicit victim x negotiator case configurations.
- **H2a (Relational Betrayal):** Same-faculty and cross-faculty betrayal cases are evaluated through explicit victim x negotiator configurations rather than a single same-group indicator.

- **H2b (Case Configuration):** Relational judgments are interpreted through explicit case contrasts such as `Hum_x_Hum`, `Hum_x_Ing`, `Hum_x_Control`, `Ing_x_Hum`, `Ing_x_Ing`, `Ing_x_Control`.
- **H3 (Moderation):** The empathy effect may vary across explicit victim x negotiator case configurations.

4 Mathematical Approach and Theoretical Foundations

The analysis employs two complementary estimators for bounded moral judgments. The primary specification is a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, moral judgments (y_{ij}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y^*_{ij}) that follows a linear relationship. Option 2 replaces isolated ingroup/outgroup indicators with explicit relational case configurations built from the paired-group structure of each judgment. The current project records the victim group first and the judged negotiator label second, so configurations such as `Hum_x_Hum`, `Hum_x_Ing`, `Hum_x_Control`, `Ing_x_Hum`, `Ing_x_Ing`, `Ing_x_Control` refer to victim x negotiator combinations rather than detached attributes. The core design vector therefore includes empathy, explicit case-configuration contrasts, and contextual conditioning terms such as role and decision context:

$$y^*_{ij} = \beta_0 + \beta_1 \mathrm{IRI}_i + \boldsymbol{\gamma}' \mathrm{CaseConfig}_{ij} + \boldsymbol{\delta}' \mathrm{Context}_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

The actual observed judgment y_{ij} relates to this latent variable via the censoring transformation:

$$y_{ij} = \max(-9, \min(9, y^*_{ij}))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

Because the Tobit model relies on a Gaussian latent-error assumption, the pipeline also fits a distribution-robust censored median specification implemented as interval-censored quantile regression ($p = 0.5$). This complementary estimator targets the conditional median of the latent bounded outcome and is less sensitive to heavy tails and non-normal disturbances:

$$Q_{0.5}(y^*_{ij} \mid \mathbf{x}_{ij}) = \mathbf{x}'_{ij}\beta_{0.5}$$

The non-parametric branch preserves the censoring structure while relaxing the parametric normality assumption. The Tobit and robustness results should therefore be interpreted jointly: Tobit provides the clustered parametric benchmark with participant-clustered standard errors by id, while the non-parametric branch first establishes a converged full-sample censored median fit and then, in the default pipeline, adds participant-level cluster bootstrap inference that resamples ids with replacement and retains all repeated observations from each sampled participant. In both branches, id is used only to account for within-participant dependence and is not a substantive explanatory variable.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. Calculate the Design Effect (Deff): As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. Determine the Effective Sample Size (ESS): The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. Inference Impact: A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors for Tobit and participant-level cluster bootstrap inference for the non-parametric robustness model after each converged full-sample fit, we ensure that both sets of p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.717, with an Effective Sample Size (ESS) of 266.9.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI latent variables.

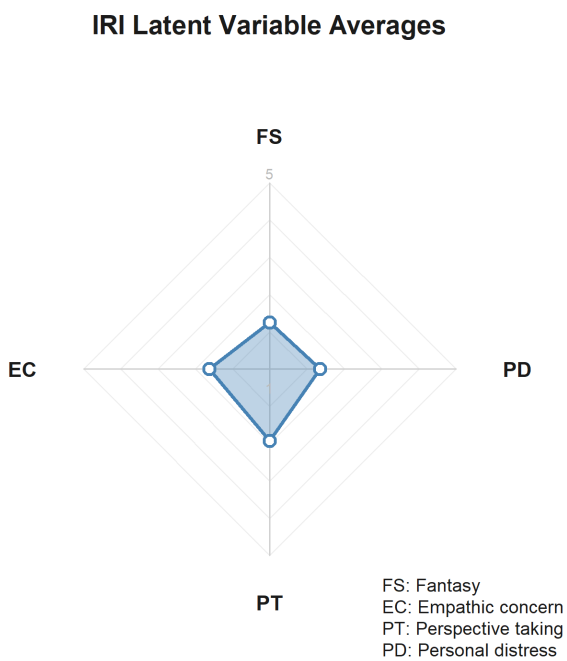

Figure 1: IRI Latent Variable Averages (Radar Plot profile).

Judgment severity distributions segmented by experimental conditions are presented in Figure 2.

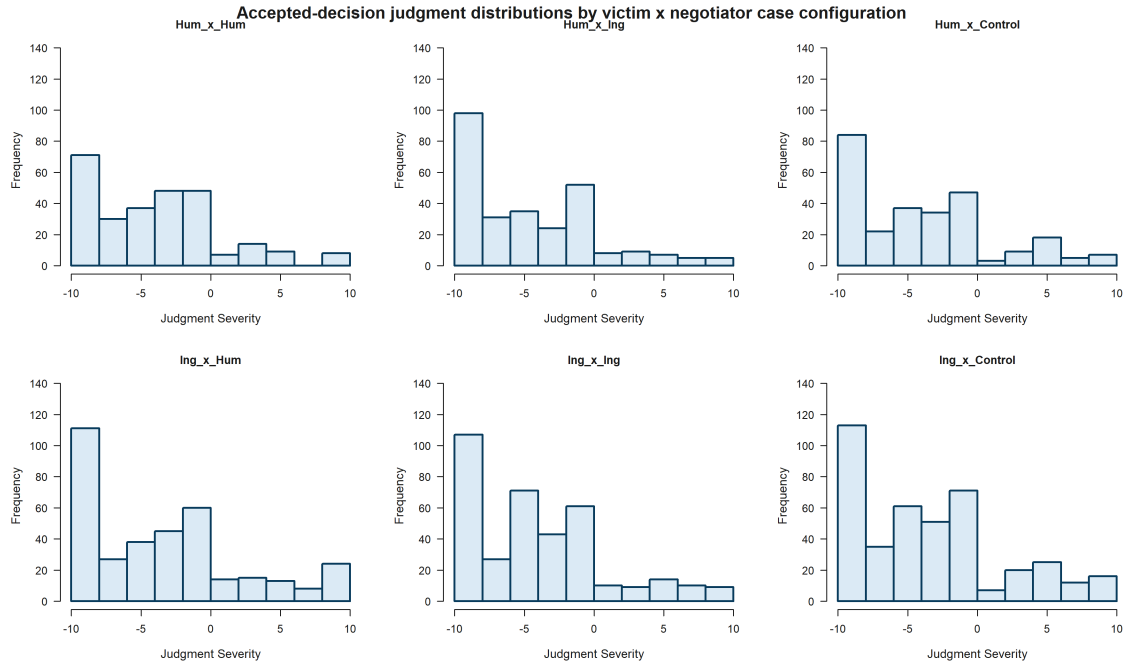

Figure 2: Accepted-decision judgment distributions by explicit victim x negotiator case configuration.

Table 2: Observed judgment summaries across explicit victim x negotiator case configurations, participant role, and decision context.

case_configuration	role	decision_accept	Observations	MeanJudgement	SDJudgement	SEJudgement	Lower95	Upper95
Hum_x_Control	observer	0	168	6.417	4.066	0.314	5.802	7.032
Hum_x_Hum	observer	0	142	6.873	3.496	0.293	6.298	7.448
Hum_x_Ing	observer	0	157	6.172	3.558	0.284	5.615	6.728
Ing_x_Control	observer	0	192	6.547	3.410	0.246	6.064	7.029
Ing_x_Hum	observer	0	169	5.757	4.510	0.347	5.077	6.437
Ing_x_Ing	observer	0	187	6.636	3.601	0.263	6.120	7.152
Hum_x_Control	victim	0	136	6.088	4.153	0.356	5.390	6.786
Hum_x_Hum	victim	0	143	5.259	4.616	0.386	4.502	6.015
Hum_x_Ing	victim	0	116	6.534	3.559	0.330	5.887	7.182
Ing_x_Control	victim	0	207	6.744	3.696	0.257	6.241	7.247
Ing_x_Hum	victim	0	215	6.637	3.891	0.265	6.117	7.157
Ing_x_Ing	victim	0	209	7.005	3.575	0.247	6.520	7.490
Hum_x_Control	observer	1	137	-3.044	5.123	0.438	-3.902	-2.186
Hum_x_Hum	observer	1	150	-3.573	4.453	0.364	-4.286	-2.861
Hum_x_Ing	observer	1	148	-3.757	4.972	0.409	-4.558	-2.956
Ing_x_Control	observer	1	173	-3.012	4.767	0.362	-3.722	-2.301
Ing_x_Hum	observer	1	178	-2.596	5.421	0.406	-3.392	-1.799
Ing_x_Ing	observer	1	189	-3.063	4.986	0.363	-3.774	-2.353
Hum_x_Control	victim	1	129	-4.054	4.931	0.434	-4.905	-3.203
Hum_x_Hum	victim	1	122	-3.434	4.846	0.439	-4.294	-2.574
Hum_x_Ing	victim	1	126	-4.825	4.427	0.394	-5.598	-4.052
Ing_x_Control	victim	1	238	-2.824	5.658	0.367	-3.542	-2.105
Ing_x_Hum	victim	1	177	-3.424	5.606	0.421	-4.250	-2.598
Ing_x_Ing	victim	1	172	-4.250	4.678	0.357	-4.949	-3.551

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 3: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Total IRI	Mean Judgment
1.000	0.500	0.554	0.509	0.830	-0.030
0.500	1.000	0.424	0.649	0.818	-0.116
0.554	0.424	1.000	0.301	0.730	0.074
0.509	0.649	0.301	1.000	0.762	-0.010
0.830	0.818	0.730	0.762	1.000	-0.029
-0.030	-0.116	0.074	-0.010	-0.029	1.000

Visual representations of these relationships are provided in Figure 3.

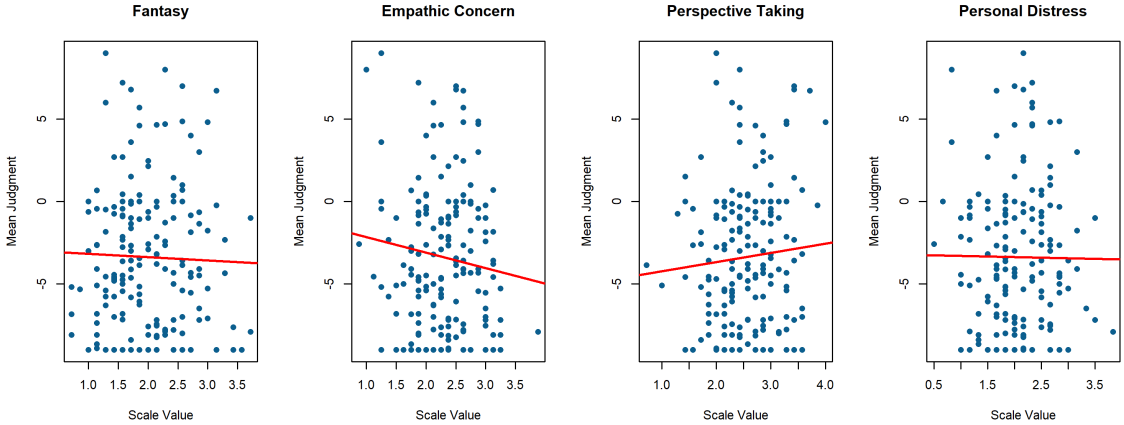

Figure 3: Bivariate Scatters: IRI Scales vs. Mean Judgment.

8 Estimator Fit Summary

The following table consolidates the fit-status information for the primary Tobit models and the non-parametric robustness branch. In the default pipeline, participant-level cluster bootstrap launches immediately after a converged full-sample non-parametric fit is available; if bootstrap is disabled manually or too few bootstrap refits converge, that status is shown explicitly.

Table 4: Estimator fit summary across Tobit and cluster-aware non-parametric specifications.

Model	Approach	Status	Converged	Iterations	BootstrapReplicates	BootstrapSuccessful	Observations	Participants	ClusterUnit
H1_A_Total_CLAD	CLAD	not_converged	FALSE	2000	1	NA	1939	199	id
H1_B_Constructs_CLAD	CLAD	failed	FALSE	NA	1	0	1939	199	id
H2a_A_Total_CLAD	CLAD	bootstrap_sparse	TRUE	46	1	1	1262	199	id
H2a_B_Constructs_CLAD	CLAD	failed	FALSE	NA	1	0	1262	199	id
H2b_A_Total_CLAD	CLAD	not_converged	FALSE	2000	1	NA	1939	199	id
H2b_B_Constructs_CLAD	CLAD	failed	FALSE	NA	1	0	1939	199	id
H3_A_Total_CLAD	CLAD	not_converged	FALSE	2000	1	NA	1939	199	id
H3_B_Constructs_CLAD	CLAD	not_converged	FALSE	2000	1	NA	1939	199	id
H1_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H1_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H2a_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1262	199	id
H2a_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1262	199	id
H2b_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H2b_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H3_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H3_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id

8.1 Hypothesis Significance Summary

The following concise table lists only hypothesis-relevant predictors that reached at least $p < 0.10$, using conventional symbols to indicate strength of evidence. If the non-parametric bootstrap is disabled, too sparse, or the censored median fit does not converge, the non-parametric column reports that status instead of inferential symbols. Dynamic figures are generated only for predictors that appear in this table with at least one significance symbol.

Table 5: Hypothesis-level significance summary across Tobit and cluster-aware non-parametric models.

Predictor	Tobit significance	Non-parametric significance
Empathy: Empathic concern is statistically significant in the Tobit model (*, p = 0.025). Figure 4 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.	*	Not
Compatibility: Interpersonal compatibility is statistically significant in the Tobit model (*, p = 0.025). Figure 5 shows that across the observed range, higher Compatibility: Interpersonal compatibility corresponds to higher predicted latent judgment.	*	Bootstrap sparse
Other Efficiency: Other efficiency is statistically significant in the Tobit model (*, p = 0.025). Figure 6 shows that across the observed range, higher Other Efficiency: Other efficiency corresponds to higher predicted latent judgment.	*	Not
Efficiency: Efficiency is statistically significant in the Tobit model (*, p = 0.025). Figure 7 shows that across the observed range, higher Efficiency: Efficiency corresponds to higher predicted latent judgment.	*	Not

8.2 Significance-Driven Figures

Only hypothesis-relevant predictors that reach $p < 0.10$ or better are visualized automatically. These dynamic figures use the saved Tobit and clustered non-parametric outputs, and participant id remains only an inference-level clustering unit rather than a substantive explanatory variable.

H1: Empathy under explicit case configuration Empathy: Empathic concern is statistically significant in the Tobit model (*, $p = 0.025$). Figure 4 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.

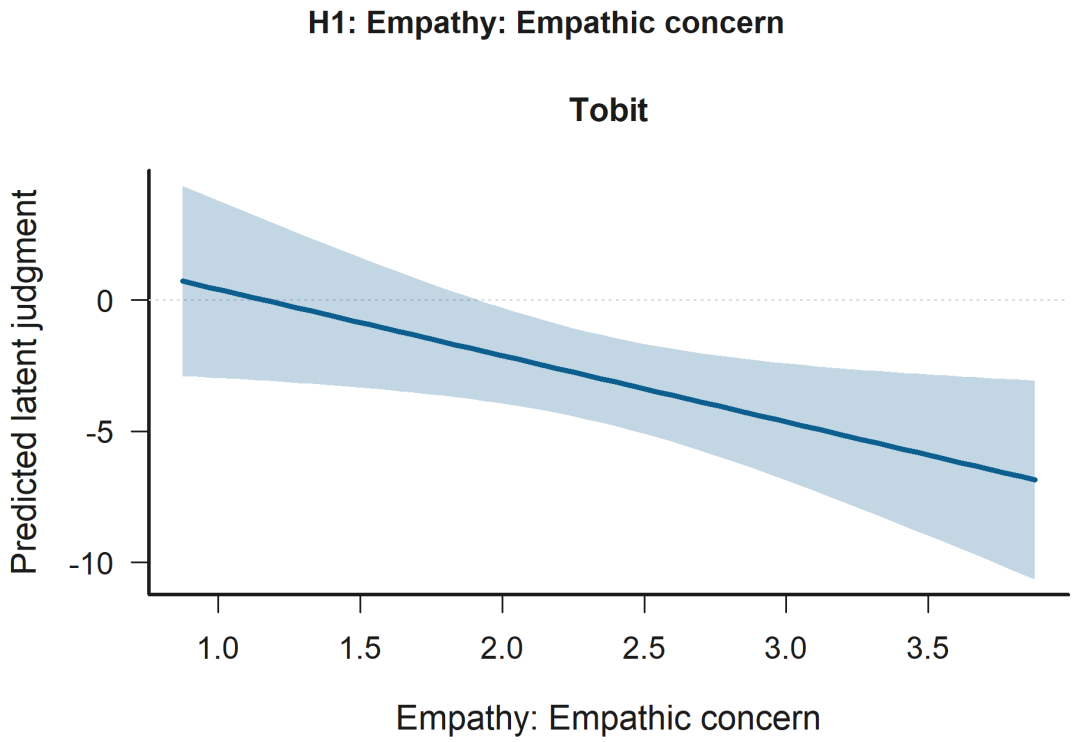

Figure 4: Effect Plot for Empathy: Empathic concern in H1: Empathy under explicit case configuration. Support comes from the Tobit model (*, p = 0.025). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in the Tobit model (*, p = 0.042). Figure 5 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

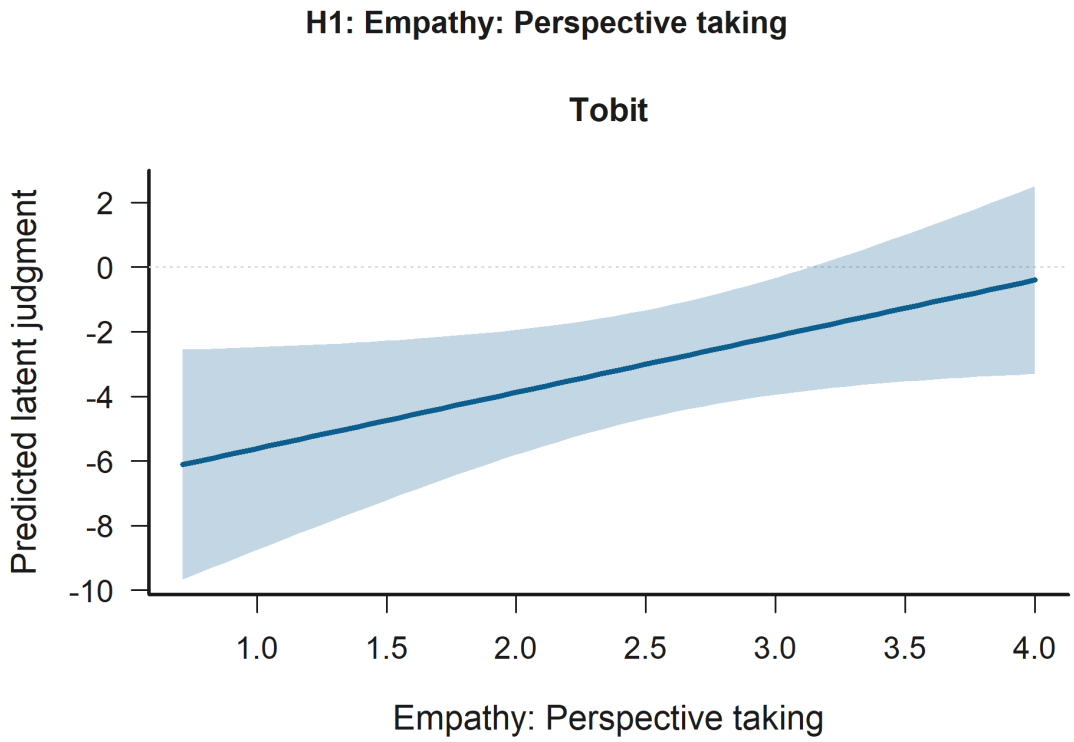

Figure 5: Effect Plot for Empathy: Perspective taking in H1: Empathy under explicit case configuration. Support comes from the Tobit model (*, p = 0.042). The panels show predicted latent judgments with 95% confidence intervals.

H2a: Relational betrayal contrasts Case configuration: Humanities victim x Engineering negotiator is statistically significant in the Tobit model (+, p = 0.061). Figure 6 shows that predicted latent judgment is lower for Humanities victim x Engineering negotiator than for All other case configurations.

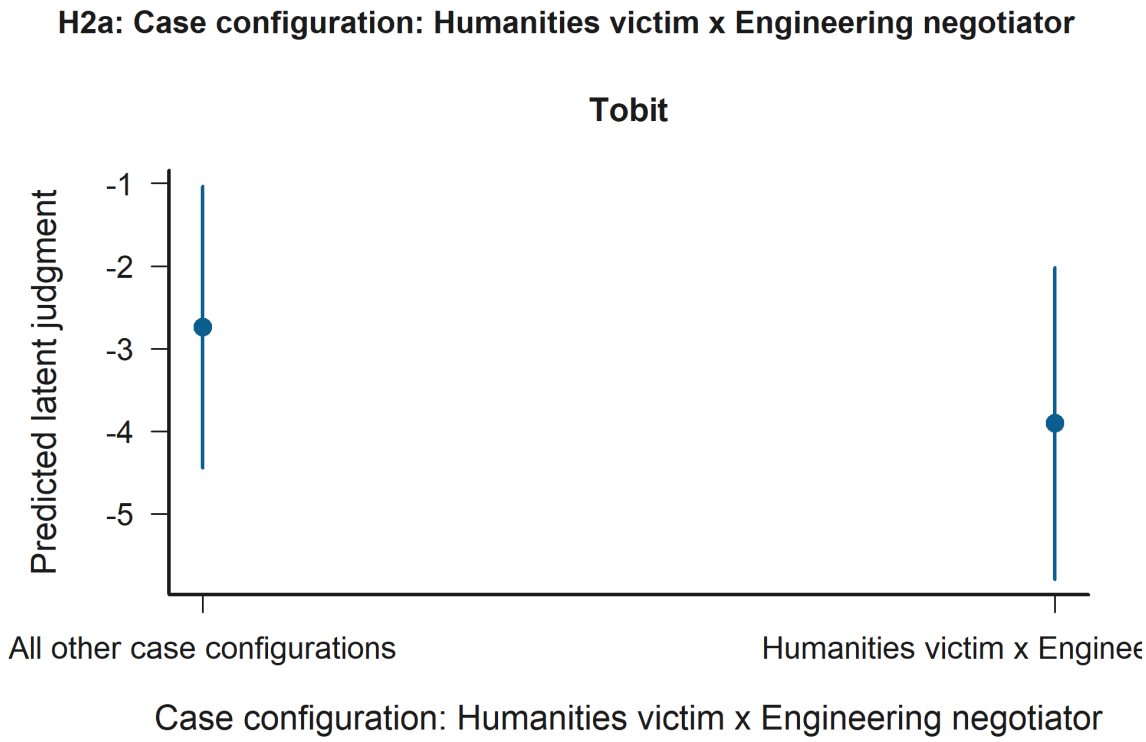

Figure 6: Grouped Prediction Plot for Case configuration: Humanities victim x Engineering negotiator in H2a: Relational betrayal contrasts. Support comes from the Tobit model (+, p = 0.061). The panels show predicted latent judgments with 95% confidence intervals.

H2b: Explicit case-configuration contrasts Case configuration: Humanities victim x Engineering negotiator is statistically significant in the Tobit model (+, p = 0.065). Figure 7 shows that predicted latent judgment is lower for Humanities victim x Engineering negotiator than for All other case configurations.

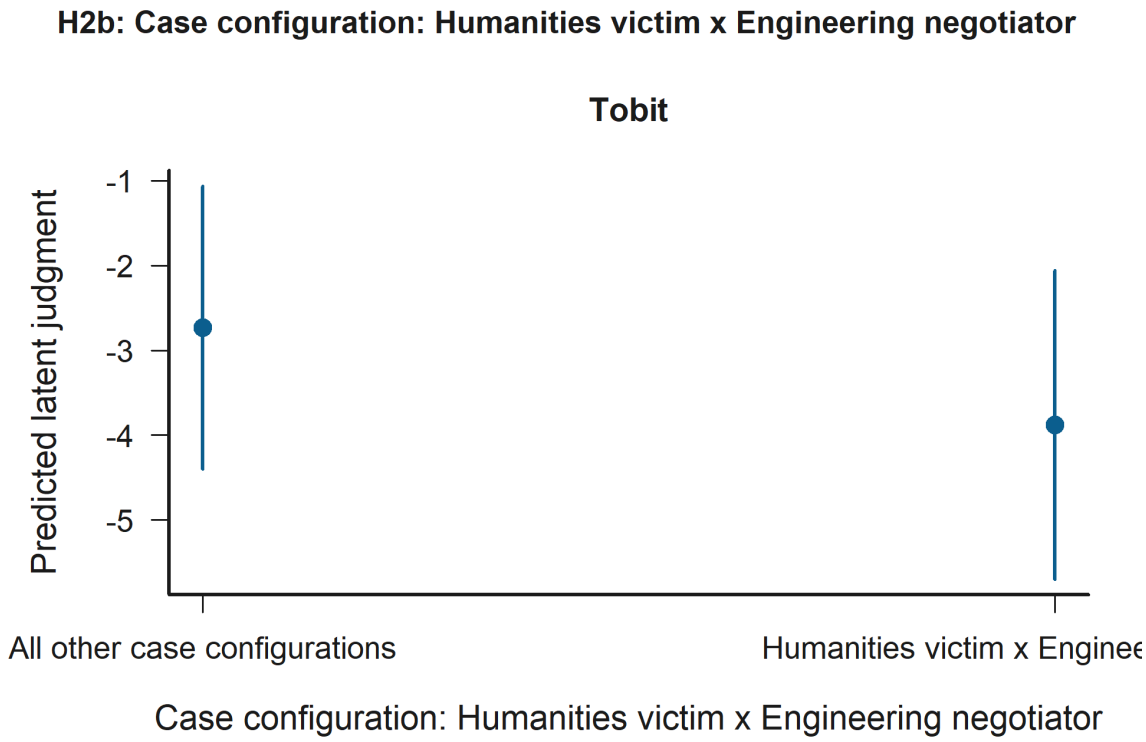

Figure 7: Grouped Prediction Plot for Case configuration: Humanities victim x Engineering negotiator in H2b: Explicit case-configuration contrasts. Support comes from the Tobit model (+, p = 0.065). The panels show predicted latent judgments with 95% confidence intervals.

H3: Empathy x case-configuration moderation Case configuration: Engineering victim x Humanities negotiator x Empathy: Empathic concern is statistically significant in the Tobit model (**, $p < 0.001$). Figure 8 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Engineering victim x Humanities negotiator.

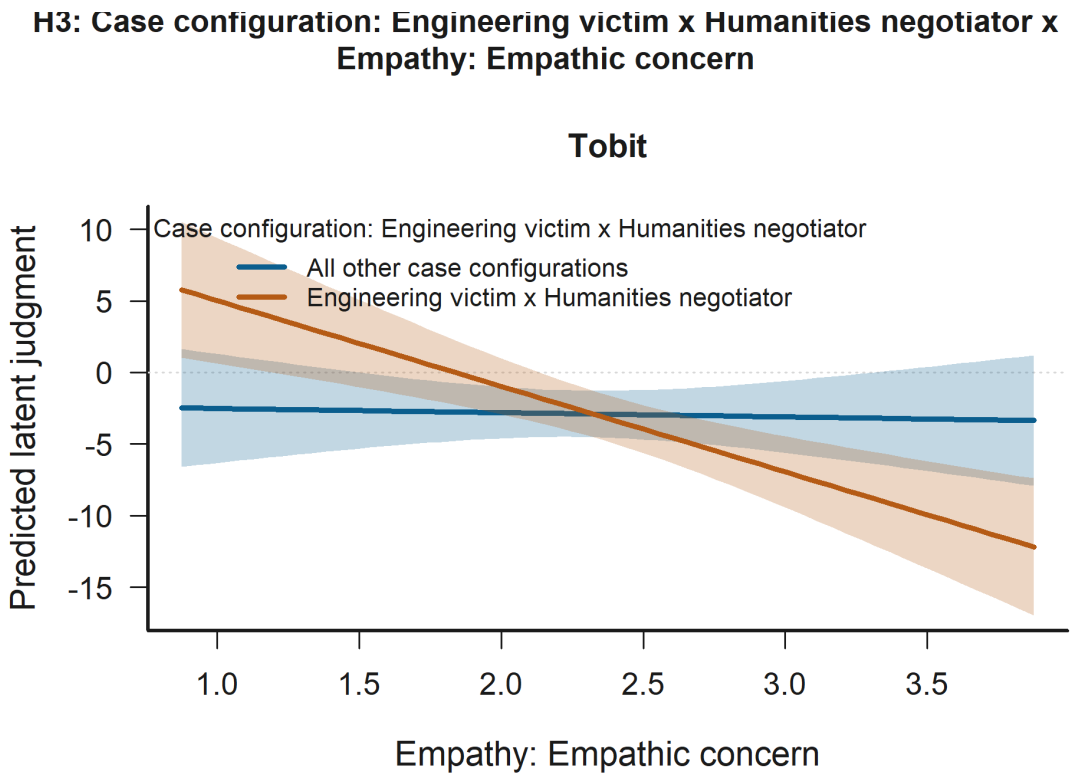

Figure 8: Interaction Plot for Case configuration: Engineering victim x Humanities negotiator x Empathy: Empathic concern in H3: Empathy x case-configuration moderation. Support comes from the Tobit model (**, $p < 0.001$). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Engineering victim x Humanities negotiator x Empathy: Personal distress is statistically significant in the Tobit model (*, $p = 0.015$). Figure 9 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Engineering victim x Humanities negotiator.

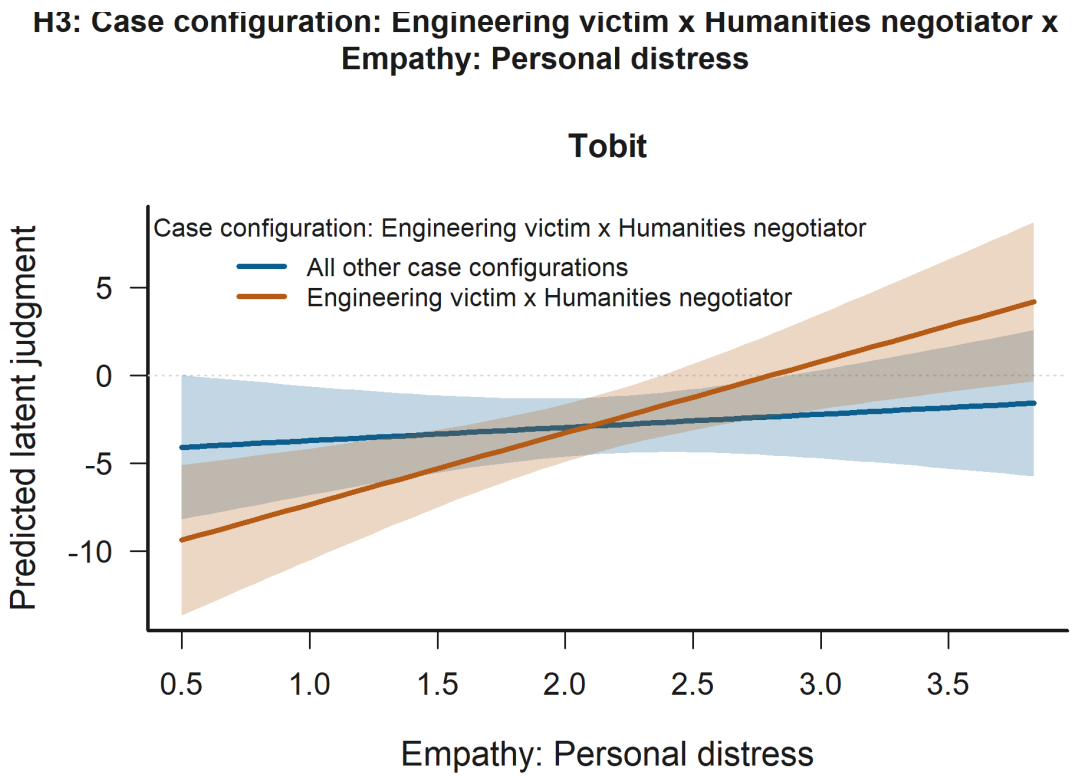

Figure 9: Interaction Plot for Case configuration: Engineering victim x Humanities negotiator x Empathy: Personal distress in H3: Empathy x case-configuration moderation. Support comes from the Tobit model (*, p = 0.015). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Humanities victim x Engineering negotiator x Empathy composite (average) is statistically significant in the Tobit model (*, p = 0.049). Figure 10 shows that the predicted relationship falls most sharply for Empathy composite (average) when the condition is Humanities victim x Engineering negotiator.

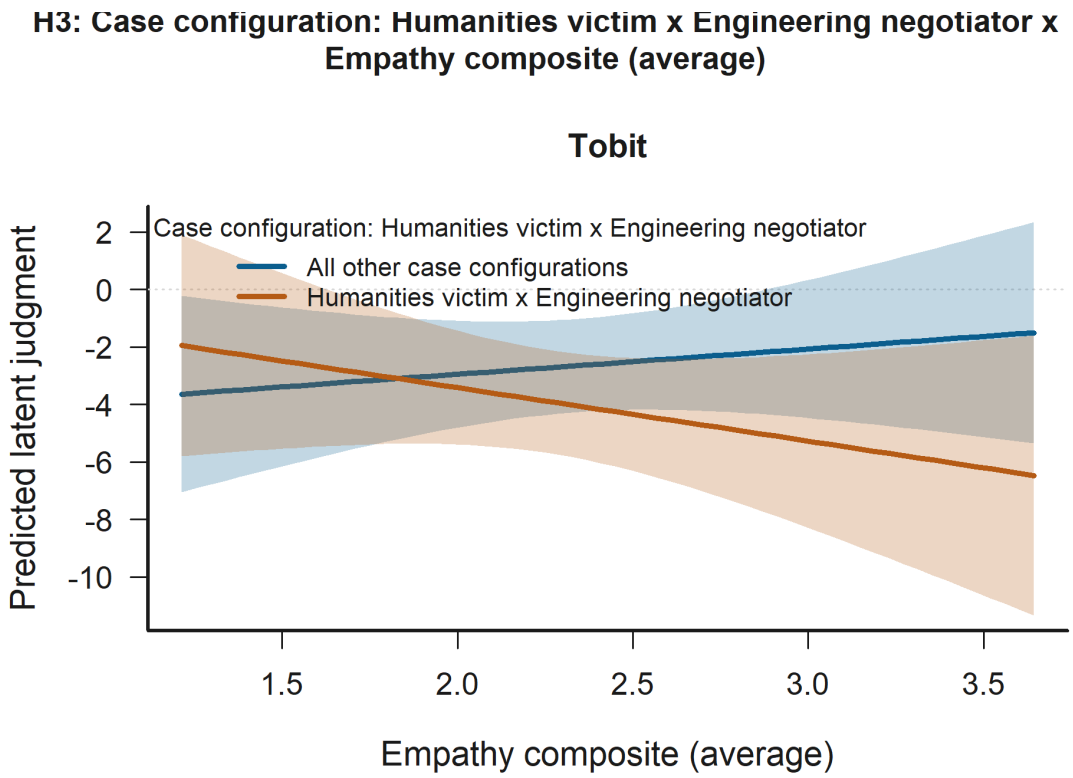

Figure 10: Interaction Plot for Case configuration: Humanities victim x Engineering negotiator x Empathy composite (average) in H3: Empathy x case-configuration moderation. Support comes from the Tobit model (*, p = 0.049). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Empathic concern is statistically significant in the Tobit model (+, p = 0.055). Figure 11 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Engineering victim x Control label hidden negotiator.

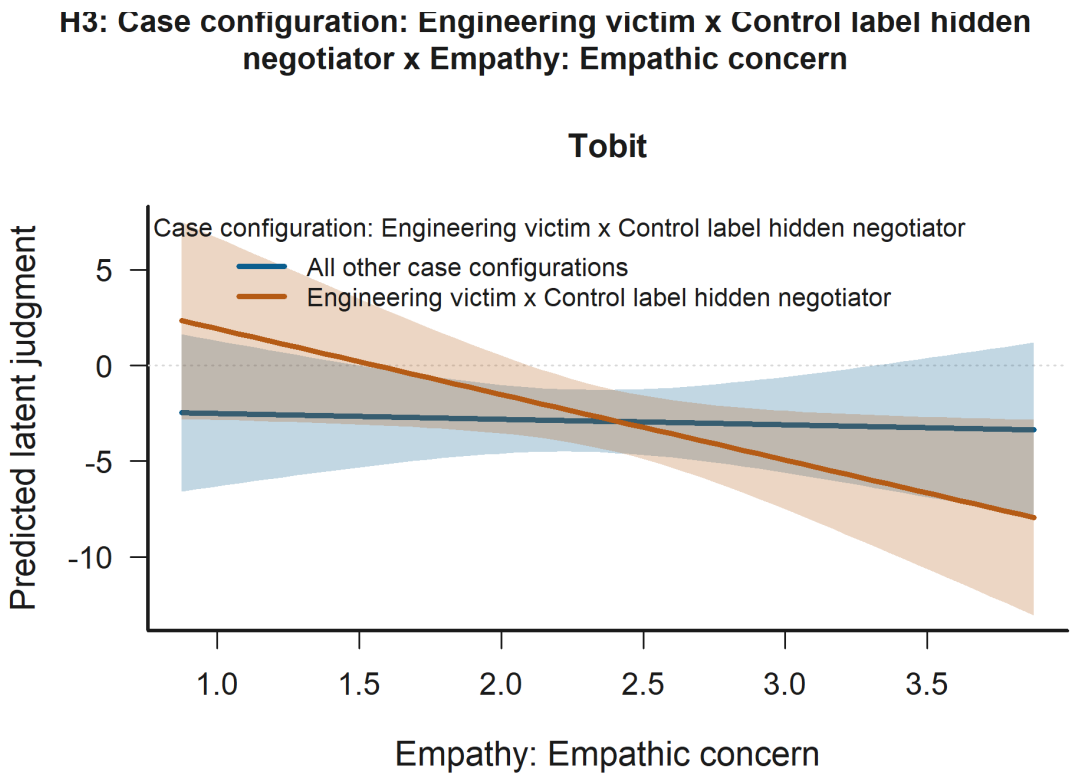

Figure 11: Interaction Plot for Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Empathic concern in H3: Empathy x case-configuration moderation. Support comes from the Tobit model (+, p = 0.055). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Engineering victim x Humanities negotiator x Empathy composite (average) is statistically significant in the Tobit model (+, p = 0.080). Figure 12 shows that the predicted relationship falls most sharply for Empathy composite (average) when the condition is Engineering victim x Humanities negotiator.

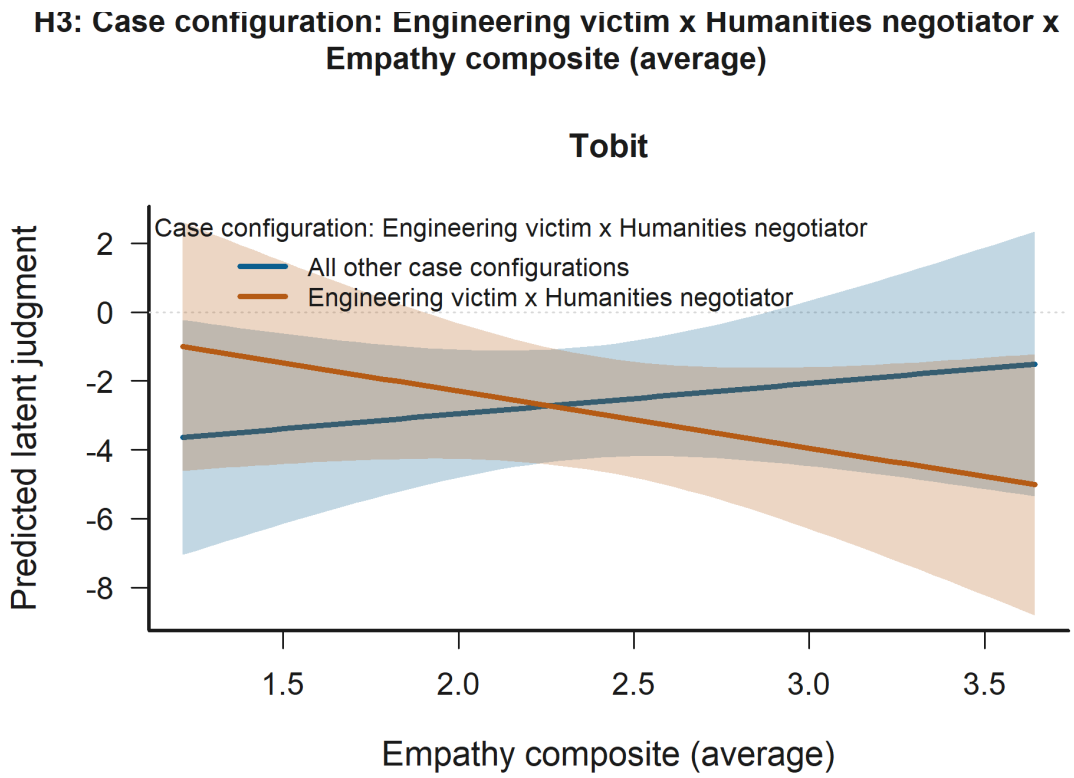

Figure 12: Interaction Plot for Case configuration: Engineering victim x Humanities negotiator x Empathy composite (average) in H3: Empathy x case-configuration moderation. Support comes from the Tobit model (+, p = 0.080). The panels show predicted latent judgments with 95% confidence intervals.

8.3 All Significant Predictors ($p < 0.10$)

The following figures extend beyond the hypothesis-target terms and visualize every predictor that reaches $p < 0.10$ in the available H1-H3 Tobit or clustered non-parametric models. This includes significant controls such as age when they clear the threshold.

Accepted-decision sample Case configuration: Engineering victim x Humanities negotiator x Empathy: Empathic concern is statistically significant in H3: Empathy x case-configuration moderation Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_case_ing_x_hum_iri_ec} shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Engineering victim x Humanities negotiator.

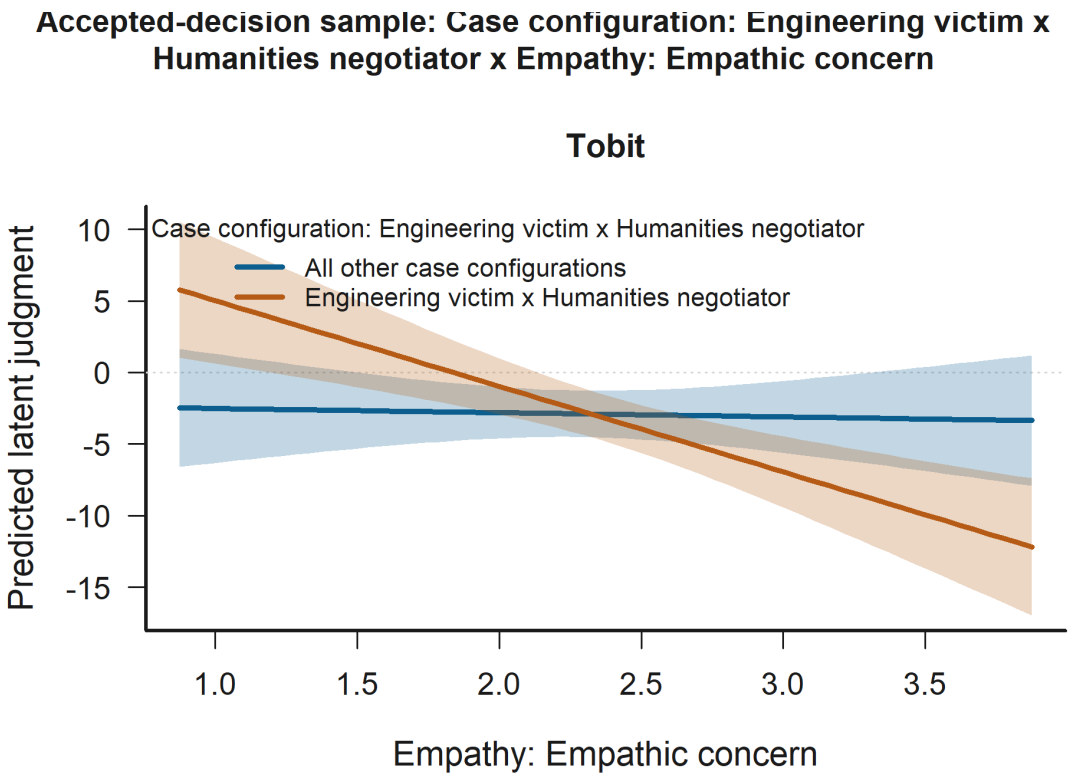

Figure 13: Interaction Plot for Case configuration: Engineering victim x Humanities negotiator x Empathy: Empathic concern in the Accepted-decision sample. Statistically significant support appears in H3: Empathy x case-configuration moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Observer role (ref = victim) is statistically significant in H2b: Explicit case-configuration contrasts Model A (Tobit), H1: Empathy under explicit case configuration Model A (Tobit), H3: Empathy x case-configuration moderation Model A (Tobit), H2b: Explicit case-configuration contrasts Model B (Tobit), H1: Empathy under explicit case configuration Model B (Tobit), and H3: Empathy x case-configuration moderation Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_role_obse shows that predicted latent judgment is higher for Observer role than for Victim role.

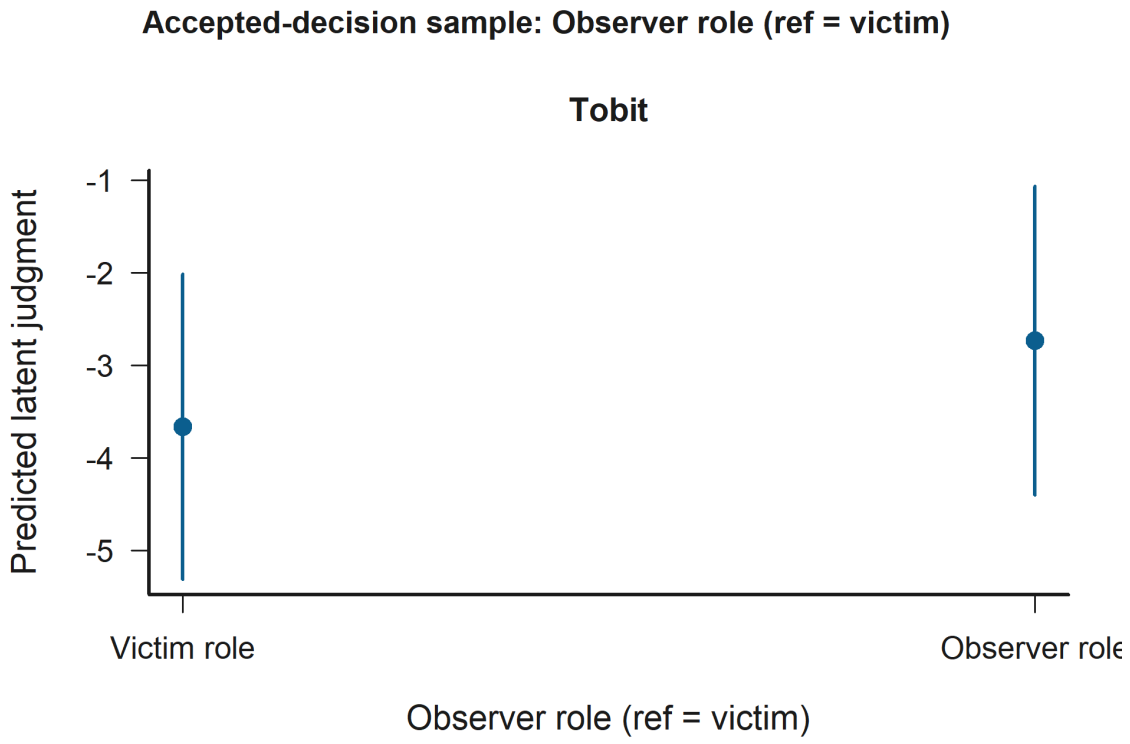

Figure 14: Grouped Prediction Plot for Observer role (ref = victim) in the Accepted-decision sample. Statistically significant support appears in H2b: Explicit case-configuration contrasts Model A (Tobit), H1: Empathy under explicit case configuration Model A (Tobit), H3: Empathy x case-configuration moderation Model A (Tobit), H2b: Explicit case-configuration contrasts Model B (Tobit), H1: Empathy under explicit case configuration Model B (Tobit), and H3: Empathy x case-configuration moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Engineering victim x Humanities negotiator x Empathy: Personal distress is statistically significant in H3: Empathy x case-configuration moderation Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_case_ing_x_hum_iri_pd} shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Engineering victim x Humanities negotiator.

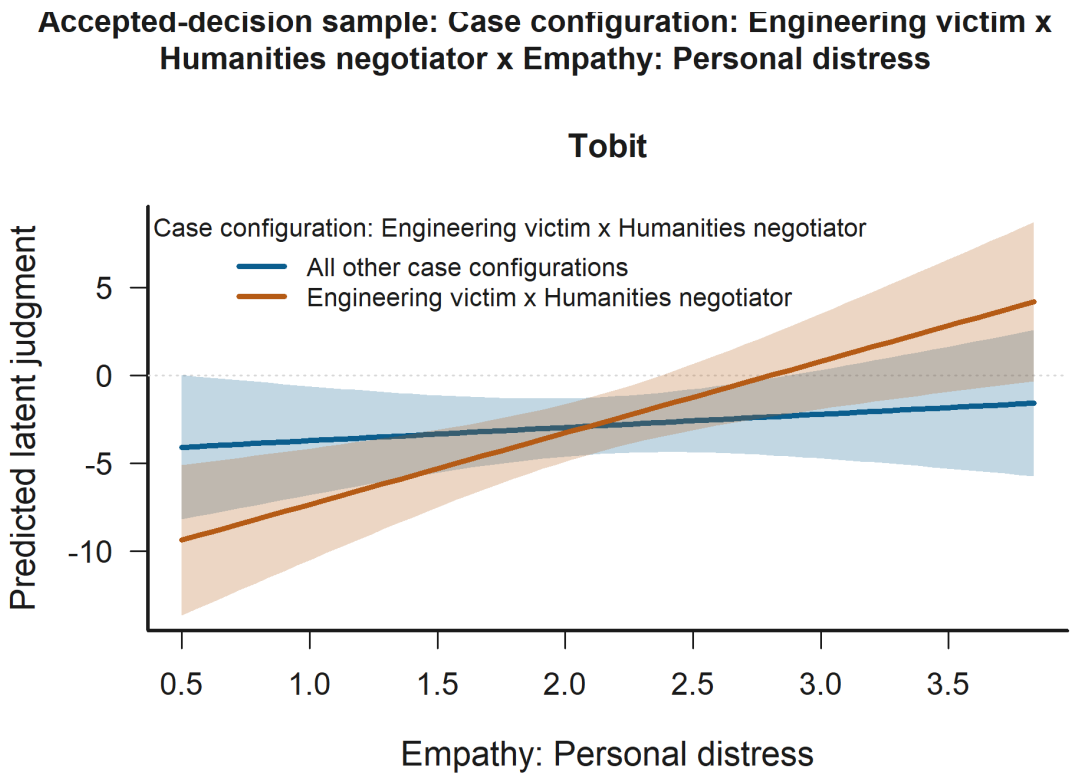

Figure 15: Interaction Plot for Case configuration: Engineering victim x Humanities negotiator x Empathy: Personal distress in the Accepted-decision sample. Statistically significant support appears in H3: Empathy x case-configuration moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in H1: Empathy under explicit case configuration Model B (Tobit) and H2b: Explicit case-configuration contrasts Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_iri_ec} shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.

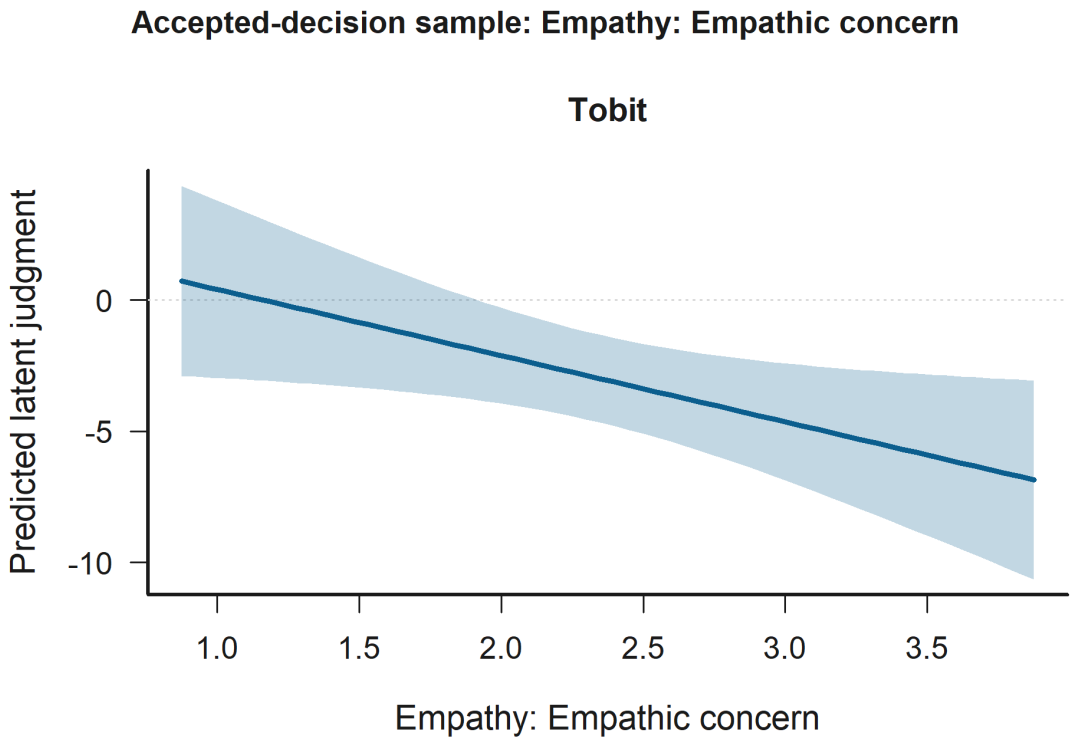

Figure 16: Effect Plot for Empathy: Empathic concern in the Accepted-decision sample. Statistically significant support appears in H1: Empathy under explicit case configuration Model B (Tobit) and H2b: Explicit case-configuration contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Age is statistically significant in H3: Empathy x case-configuration moderation Model A (Tobit), H1: Empathy under explicit case configuration Model A (Tobit), H2b: Explicit case-configuration contrasts Model A (Tobit), H2b: Explicit case-configuration contrasts Model B (Tobit), H1: Empathy under explicit case configuration Model B (Tobit), and H3: Empathy x case-configuration moderation Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_age} shows that across the observed range, higher Age corresponds to higher predicted latent judgment.

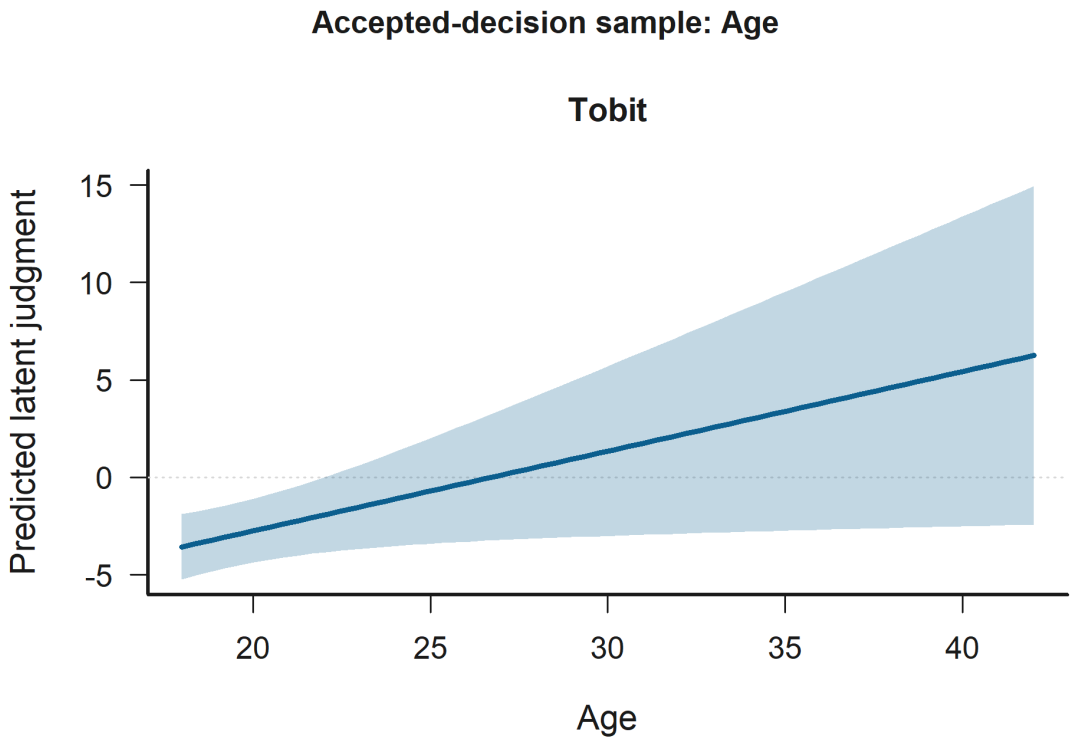

Figure 17: Effect Plot for Age in the Accepted-decision sample. Statistically significant support appears in H3: Empathy x case-configuration moderation Model A (Tobit), H1: Empathy under explicit case configuration Model A (Tobit), H2b: Explicit case-configuration contrasts Model A (Tobit), H2b: Explicit case-configuration contrasts Model B (Tobit), H1: Empathy under explicit case configuration Model B (Tobit), and H3: Empathy x case-configuration moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in H2b: Explicit case-configuration contrasts Model B (Tobit) and H1: Empathy under explicit case configuration Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_iri_pt} shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

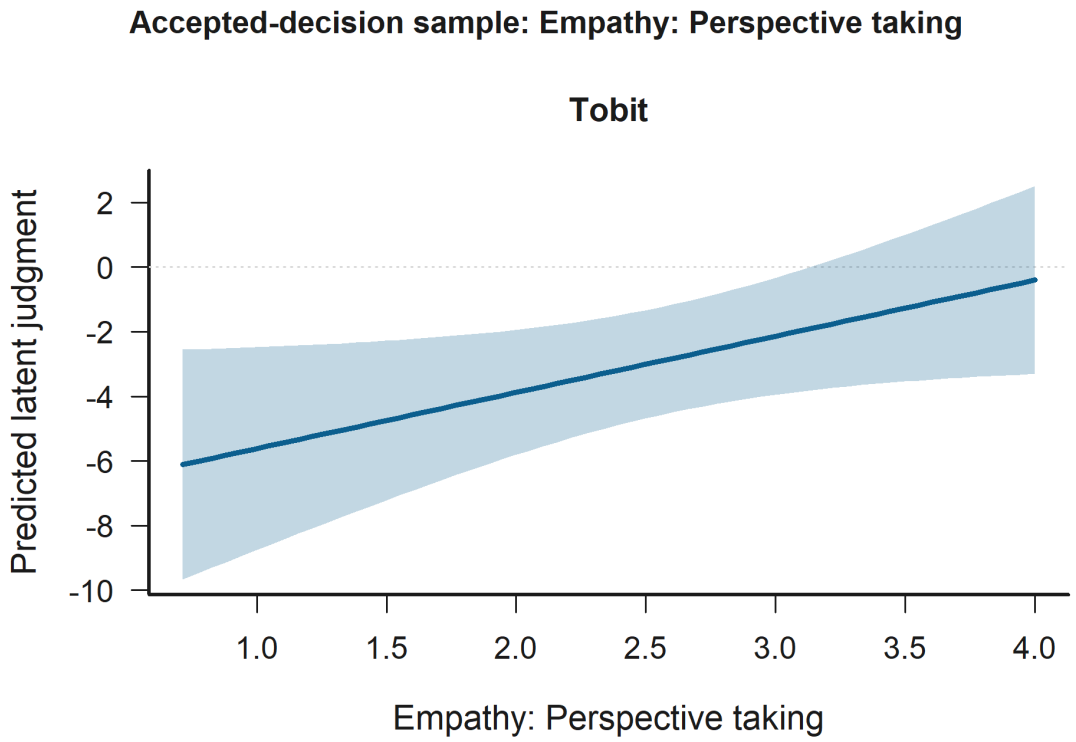

Figure 18: Effect Plot for Empathy: Perspective taking in the Accepted-decision sample. Statistically significant support appears in H2b: Explicit case-configuration contrasts Model B (Tobit) and H1: Empathy under explicit case configuration Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Humanities victim x Engineering negotiator x Empathy composite (average) is statistically significant in H3: Empathy x case-configuration moderation Model A (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_case_hum_x_ing_iri_total} shows that the predicted relationship falls most sharply for Empathy composite (average) when the condition is Humanities victim x Engineering negotiator.

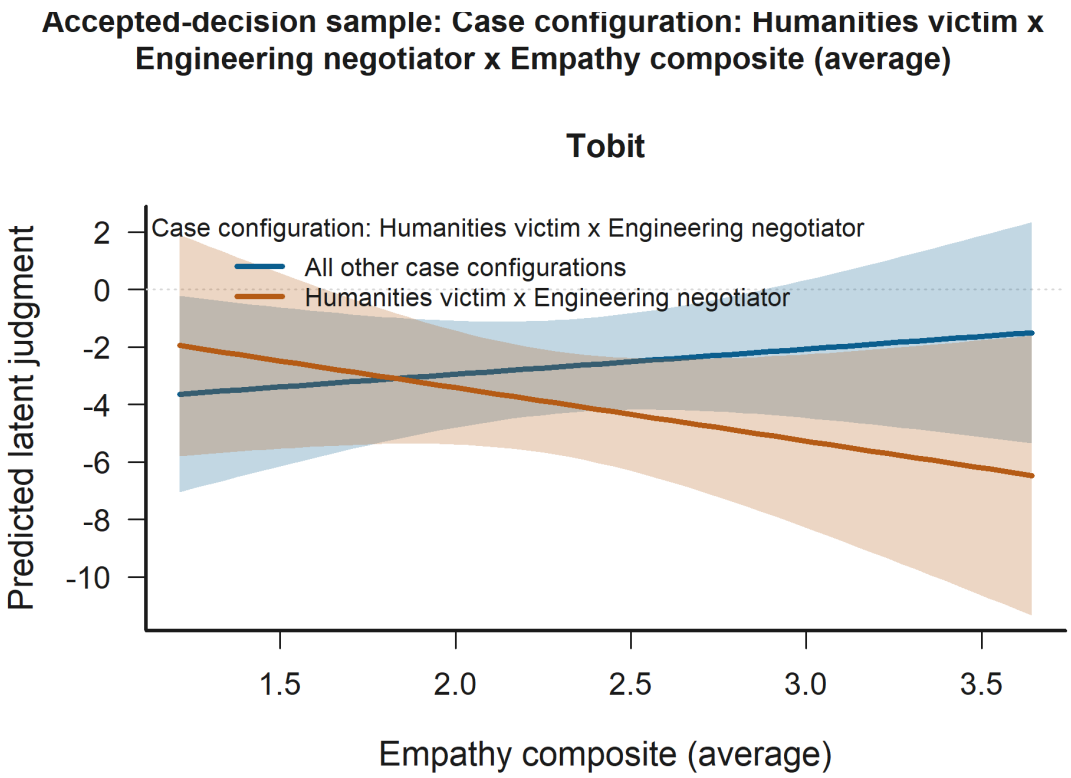

Figure 19: Interaction Plot for Case configuration: Humanities victim x Engineering negotiator x Empathy composite (average) in the Accepted-decision sample. Statistically significant support appears in H3: Empathy x case-configuration moderation Model A (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Empathic concern is statistically significant in H3: Empathy x case-configuration moderation Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_case_ing_x_control_iri_ec} shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Engineering victim x Control label hidden negotiator.

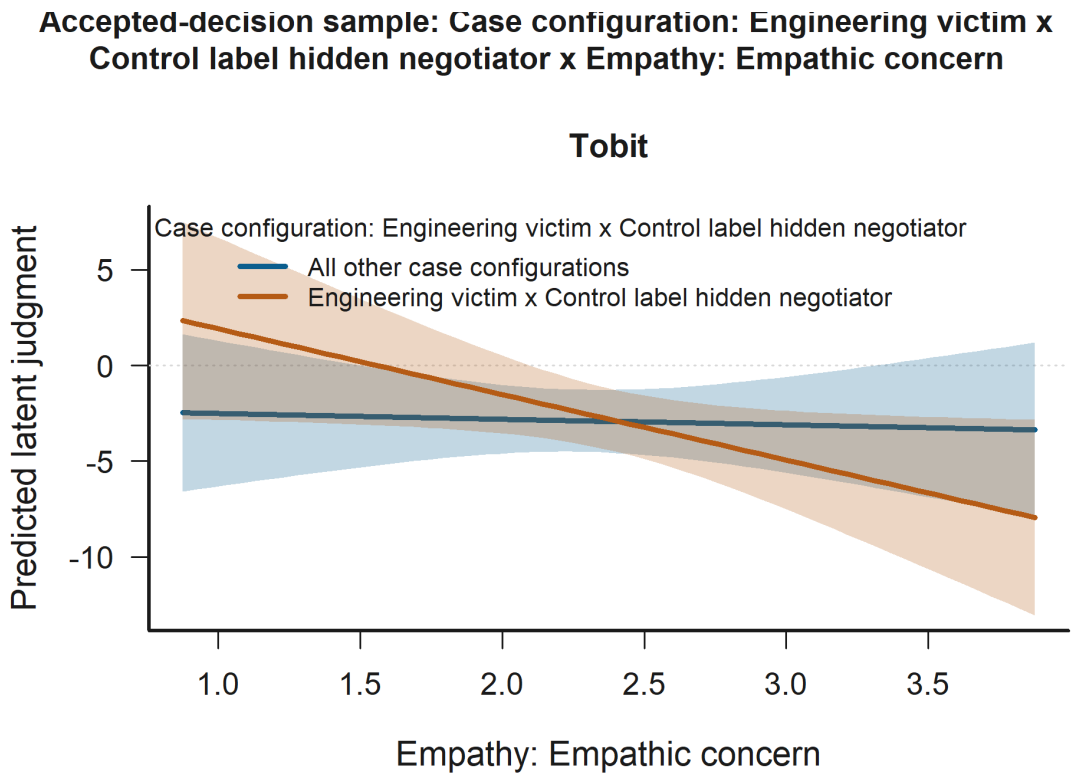

Figure 20: Interaction Plot for Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Empathic concern in the Accepted-decision sample. Statistically significant support appears in H3: Empathy x case-configuration moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Humanities victim x Engineering negotiator is statistically significant in H1: Empathy under explicit case configuration Model A (Tobit), H2b: Explicit case-configuration contrasts Model A (Tobit), H1: Empathy under explicit case configuration Model B (Tobit), and H2b: Explicit case-configuration contrasts Model B (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_case_ shows that predicted latent judgment is lower for Humanities victim x Engineering negotiator than for All other case configurations.

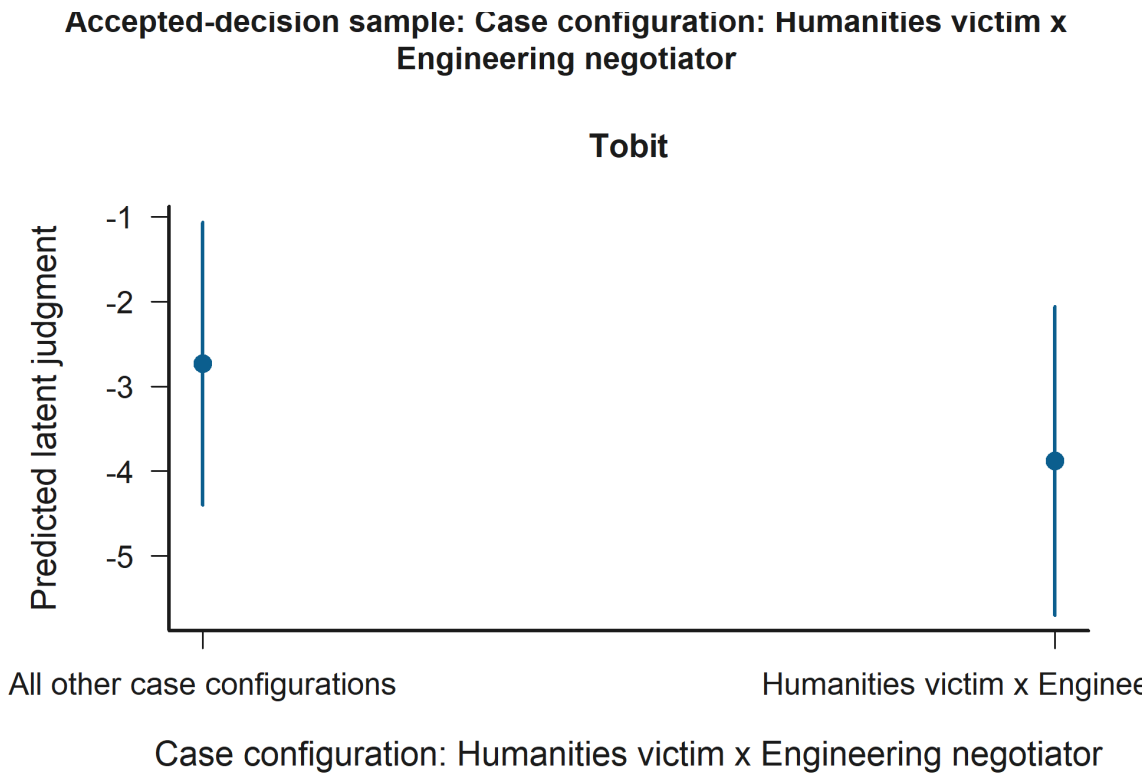

Figure 21: Grouped Prediction Plot for Case configuration: Humanities victim x Engineering negotiator in the Accepted-decision sample. Statistically significant support appears in H1: Empathy under explicit case configuration Model A (Tobit), H2b: Explicit case-configuration contrasts Model A (Tobit), H1: Empathy under explicit case configuration Model B (Tobit), and H2b: Explicit case-configuration contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Engineering victim x Humanities negotiator x Empathy composite (average) is statistically significant in H3: Empathy x case-configuration moderation Model A (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_case_ing_x_hum_iri_total} shows that the predicted relationship falls most sharply for Empathy composite (average) when the condition is Engineering victim x Humanities negotiator.

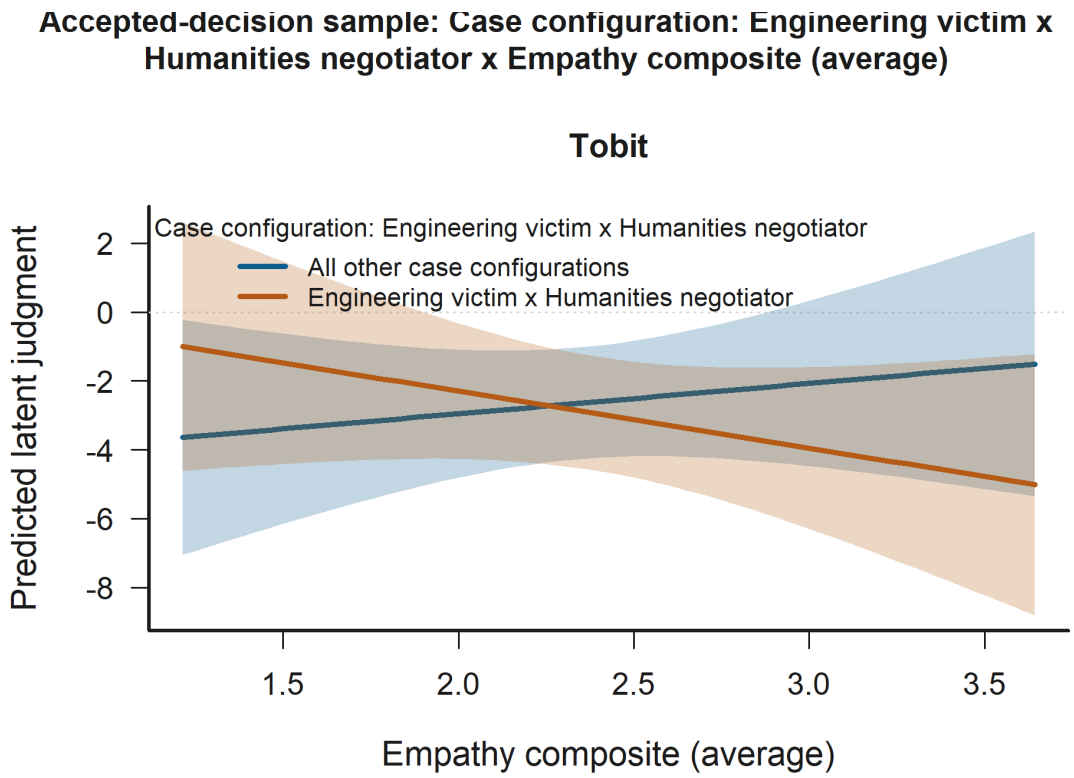

Figure 22: Interaction Plot for Case configuration: Engineering victim x Humanities negotiator x Empathy composite (average) in the Accepted-decision sample. Statistically significant support appears in H3: Empathy x case-configuration moderation Model A (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Engineering victim x Humanities negotiator is statistically significant in H3: Empathy x case-configuration moderation Model A (Tobit). Figure \{}ref{fig:sig_all_accepted_decision_sample_ shows that predicted latent judgment is higher for Engineering victim x Humanities negotiator than for All other case configurations.

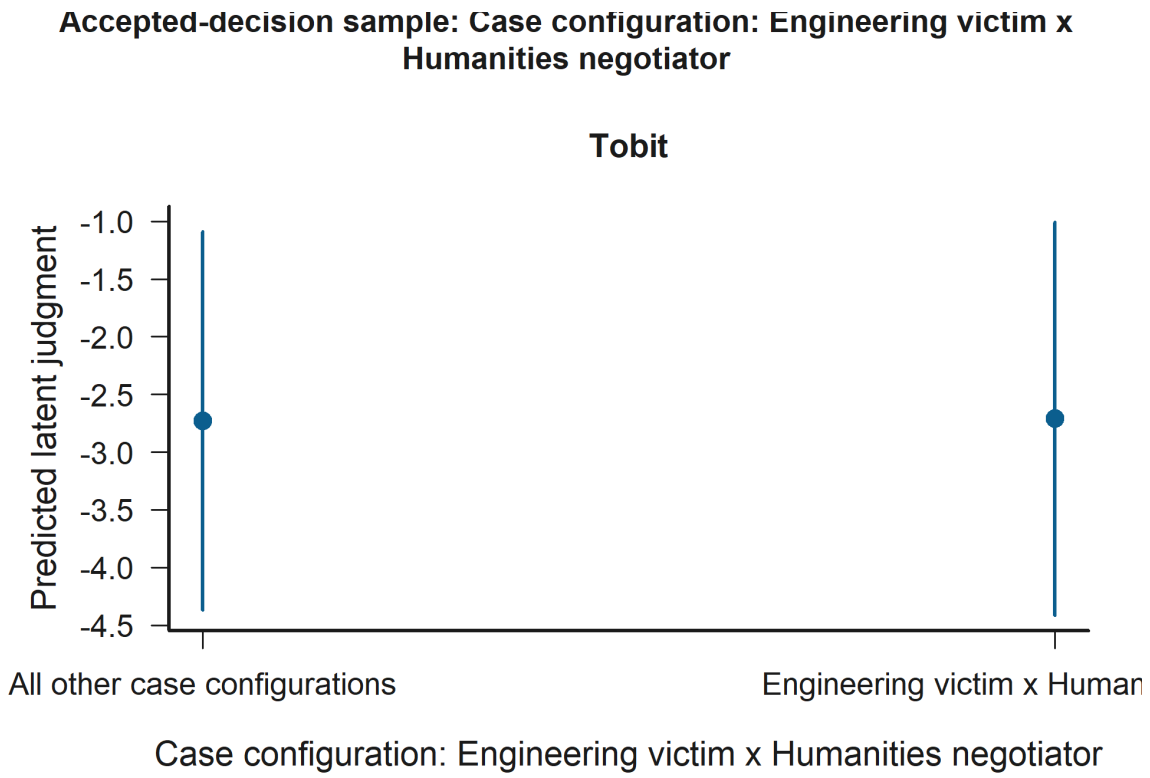

Figure 23: Grouped Prediction Plot for Case configuration: Engineering victim x Humanities negotiator in the Accepted-decision sample. Statistically significant support appears in H3: Empathy x case-configuration moderation Model A (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Betrayal sample Observer role (ref = victim) is statistically significant in H2a: Relational betrayal contrasts Model A (Tobit) and H2a: Relational betrayal contrasts Model B (Tobit). Figure \{}ref{fig:sig_all_betrayal_sample_role_observer} shows that predicted latent judgment is higher for Observer role than for Victim role.

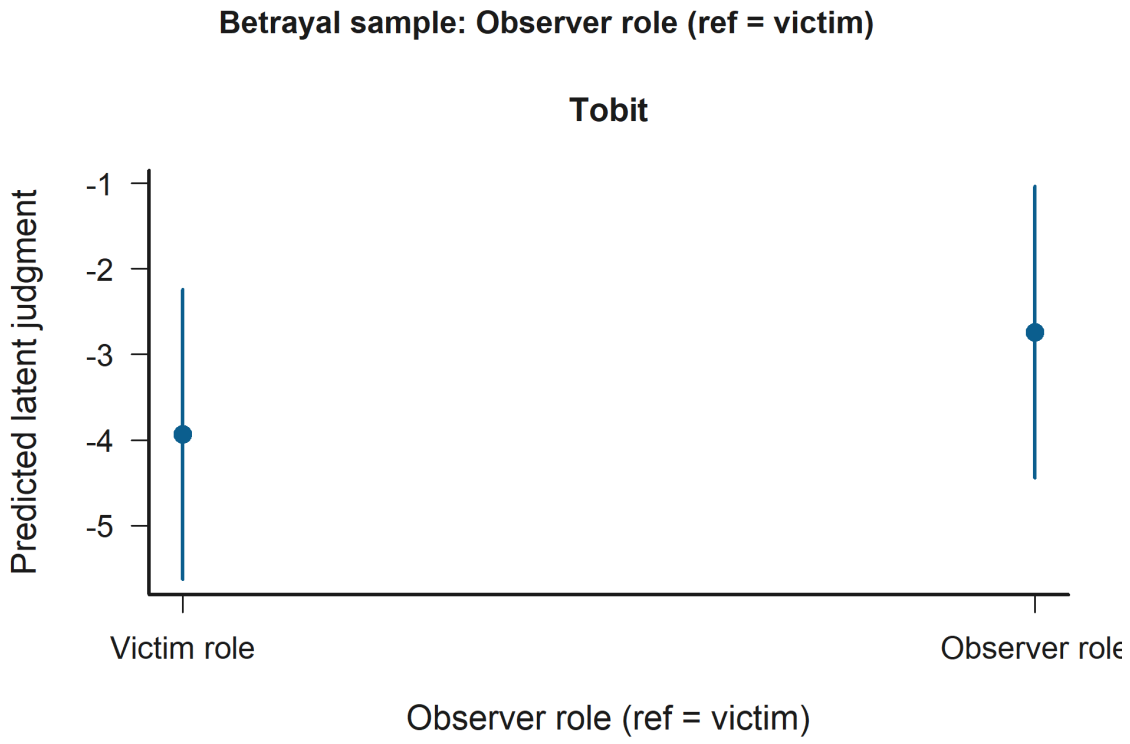

Figure 24: Grouped Prediction Plot for Observer role (ref = victim) in the Betrayal sample. Statistically significant support appears in H2a: Relational betrayal contrasts Model A (Tobit) and H2a: Relational betrayal contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in H2a: Relational betrayal contrasts Model B (Tobit). Figure \{}ref{fig:sig_all_betrayal_sample_iri_ec} shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.

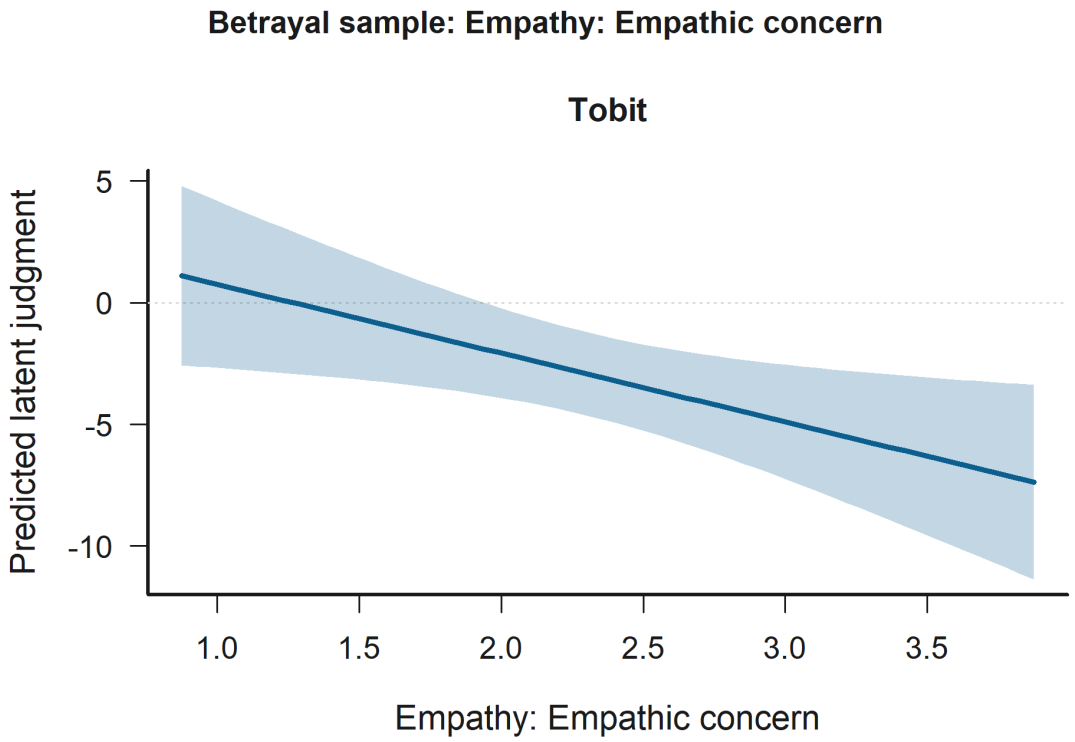

Figure 25: Effect Plot for Empathy: Empathic concern in the Betrayal sample. Statistically significant support appears in H2a: Relational betrayal contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Age is statistically significant in H2a: Relational betrayal contrasts Model A (Tobit) and H2a: Relational betrayal contrasts Model B (Tobit). Figure \{}ref{fig:sig_all_betrayal_sample_age} shows that across the observed range, higher Age corresponds to higher predicted latent judgment.

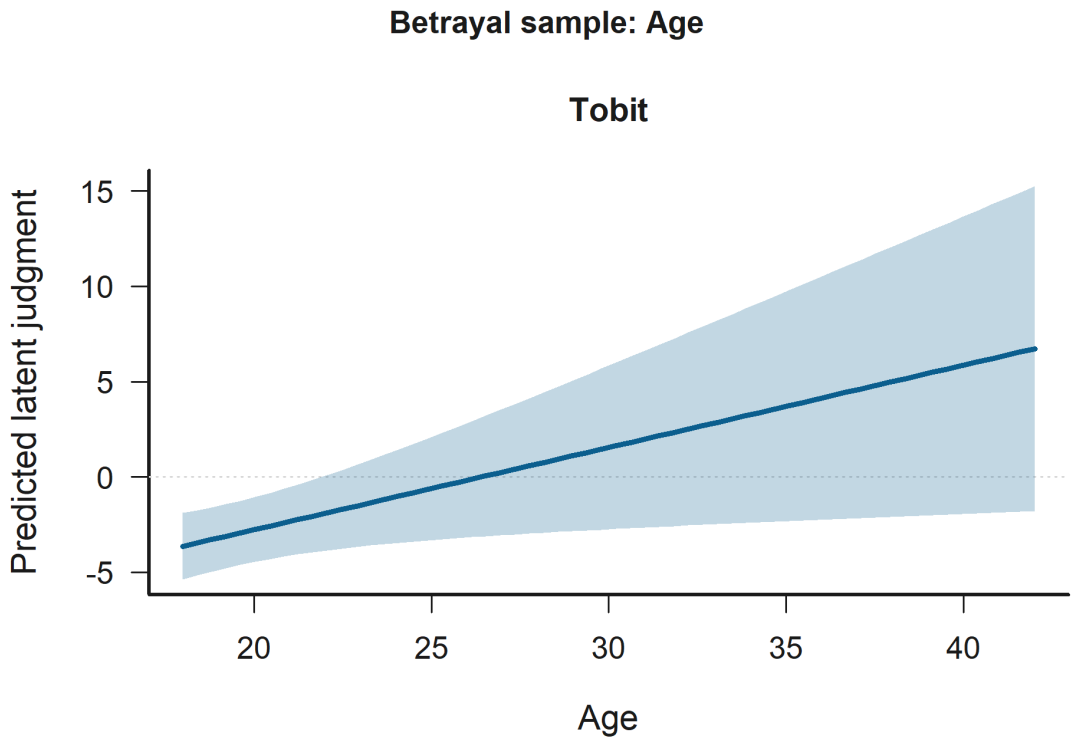

Figure 26: Effect Plot for Age in the Betrayal sample. Statistically significant support appears in H2a: Relational betrayal contrasts Model A (Tobit) and H2a: Relational betrayal contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Case configuration: Humanities victim x Engineering negotiator is statistically significant in H2a: Relational betrayal contrasts Model A (Tobit) and H2a: Relational betrayal contrasts Model B (Tobit). Figure \{}ref{fig:sig_all_betrayal_sample_case_hum_x_ing} shows that predicted latent judgment is lower for Humanities victim x Engineering negotiator than for All other case configurations.

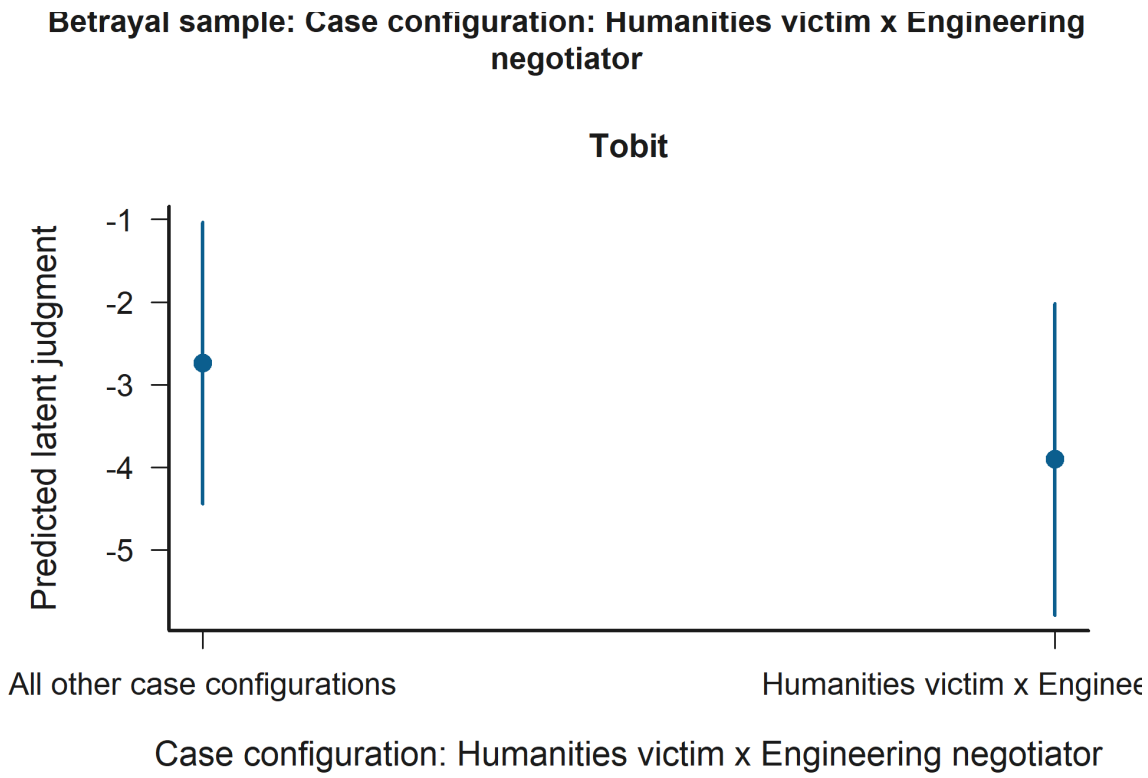

Figure 27: Grouped Prediction Plot for Case configuration: Humanities victim x Engineering negotiator in the Betrayal sample. Statistically significant support appears in H2a: Relational betrayal contrasts Model A (Tobit) and H2a: Relational betrayal contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in H2a: Relational betrayal contrasts Model B (Tobit). Figure \{}ref{fig:sig_all_betrayal_sample_iri_pt} shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

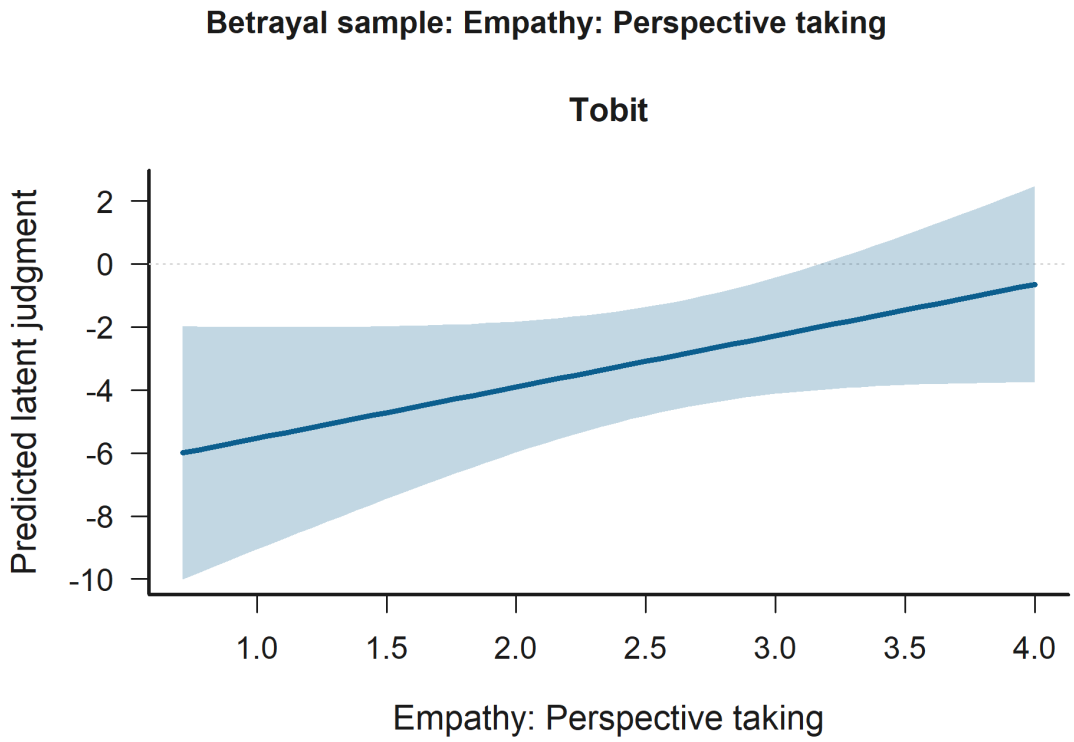

Figure 28: Effect Plot for Empathy: Perspective taking in the Betrayal sample. Statistically significant support appears in H2a: Relational betrayal contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

8.4 Integrated Hypothesis Conclusions

The following summary restates each original hypothesis and indicates whether the available Tobit estimates and the cluster-aware non-parametric models support it in the current data. Non-parametric conclusions are drawn when the participant-level bootstrap inference is available and are otherwise labeled explicitly.

- H1. Original hypothesis: Higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on explicit victim x negotiator case configurations. Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is negative but not statistically significant ($p = 0.503$). Model B supports the hypothesis through Empathy: Empathic concern with a negative association ($p = 0.025$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.004$) and Age with a positive association ($p = 0.035$). Non-parametric conclusion: no converged second-phase non-parametric model is available, so the robustness check is inconclusive for this hypothesis.

- H2a. Original hypothesis: Same-faculty and cross-faculty betrayal cases are evaluated through explicit victim x negotiator configurations rather than a single same_group_harm flag. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; none of the betrayal-sample case-configuration contrasts are statistically significant, and the closest signal is Case configuration: Humanities victim x Engineering negotiator

with a negative association ($p = 0.061$). Model B does not support the hypothesis; none of the betrayal-sample case-configuration contrasts are statistically significant, and the closest signal is Case configuration: Humanities victim x Engineering negotiator with a negative association ($p = 0.097$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.002$) and Empathy: Empathic concern with a negative association ($p = 0.016$). Non-parametric conclusion: the full-sample non-parametric fit converged, but fewer than two participant-level bootstrap refits succeeded, so the robustness check is not interpreted inferentially for this hypothesis.

- H2b. Original hypothesis: Relational judgments are interpreted through explicit victim x negotiator case configurations such as Hum_x_Ing, Hum_x_Control, Ing_x_Hum, Ing_x_Ing, and Ing_x_Control. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; none of the accepted-sample case-configuration contrasts are statistically significant, and the closest signal is Case configuration: Humanities victim x Engineering negotiator with a negative association ($p = 0.065$). Model B does not support the hypothesis; none of the accepted-sample case-configuration contrasts are statistically significant, and the closest signal is Case configuration: Humanities victim x Engineering negotiator with a negative association ($p = 0.095$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.004$) and Empathy: Empathic concern with a negative association ($p = 0.025$). Non-parametric conclusion: no converged second-phase non-parametric model is available, so the robustness check is inconclusive for this hypothesis.

- H3. Original hypothesis: The empathy effect may vary across explicit victim x negotiator pairings, so moderation is modeled through empathy interactions with case-configuration contrasts. Tobit conclusion: the available models support the hypothesis. Model A supports the hypothesis through Case configuration: Humanities victim x Engineering negotiator x Empathy composite (average) with a negative association ($p = 0.049$). Model B supports the hypothesis through Case configuration: Engineering victim x Humanities negotiator x Empathy: Empathic concern with a negative association ($p < 0.001$) and Case configuration: Engineering victim x Humanities negotiator x Empathy: Personal distress with a positive association ($p = 0.015$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.004$) and Age with a positive association ($p = 0.032$). Non-parametric conclusion: no converged second-phase non-parametric model is available, so the robustness check is inconclusive for this hypothesis.

9 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model, coupled with non-parametric robustness outputs, natural language interpretive narratives, and estimator-specific diagnostics, are provided below. In the default pipeline, converged non-parametric fits immediately attempt participant-level cluster-bootstrap inference; the report labels deferred and sparse-bootstrap cases explicitly when full inference is not available.

9.1 H1: Empathy Effect

This section evaluates the primary effect of empathy on moral judgments while holding explicit victim x negotiator case configurations constant.

9.1.1 Model A: Composite Empathy

Table 6: H1 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.201	4.407	0.003
Empathy composite (average)	-0.671	1.002	0.503
Case configuration: Humanities victim x Engineering negotiator	-1.145	0.621	0.065
Case configuration: Humanities victim x Control label hidden negotiator	-0.050	0.551	0.928
Case configuration: Engineering victim x Humanities negotiator	0.013	0.641	0.983
Case configuration: Engineering victim x Engineering negotiator	-0.757	0.564	0.179
Case configuration: Engineering victim x Control label hidden negotiator	0.317	0.554	0.566
Observer role (ref = victim)	0.929	0.324	0.004
Engineering participant (ref = humanities)	1.415	0.889	0.111
Man (ref = woman)	0.356	0.892	0.690
Age	0.408	0.193	0.035
Socioeconomic status	0.267	0.392	0.495
Negotiator 2 (ref = negotiator 1)	0.621	0.261	0.017
Scale variance (log)	1.891	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.201 (p=0.003). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.671. The p-value of 0.503 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -1.145. The p-value of 0.065 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of -0.050. The p-value of 0.928 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 0.013. The p-value of 0.983 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.757. The p-value of 0.179 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 0.317. The p-value of 0.566 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.929. The p-value of 0.004 indicates statistically significant, meaning the predictor

has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.415. The p-value of 0.111 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.356. The p-value of 0.690 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.408. The p-value of 0.035 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.267. The p-value of 0.495 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.891, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.954, p = 3.779e-24). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 7: H1 Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-15.801	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	-0.447	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator	-0.892	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator	-0.164	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator	-0.147	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator	-0.418	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator	0.090	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.999	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.129	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.175	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.615	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.012	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.271	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.801 (p=NA). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.447. The p-value of NA the p-value cannot be determined. The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -0.892. The p-value of NA the p-value cannot be determined. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of -0.164. The p-value of NA the p-value cannot be determined. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of -0.147. The p-value of NA the p-value cannot be determined. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotia-

tor case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.418. The p-value of NA the p-value cannot be determined. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 0.090. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.999. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.129. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.175. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.615. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.012. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.1.2 Model B: Separated Empathy Constructs

Table 8: H1 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-12.864	4.619	0.005
Empathy: Fantasy scale	-0.930	0.804	0.247
Empathy: Empathic concern	-2.523	1.123	0.025
Empathy: Perspective taking	1.738	0.857	0.042
Empathy: Personal distress	1.291	0.965	0.181
Case configuration: Humanities victim x Engineering negotiator	-1.030	0.617	0.095
Case configuration: Humanities victim x Control label hidden negotiator	-0.094	0.552	0.865
Case configuration: Engineering victim x Humanities negotiator	0.071	0.639	0.912
Case configuration: Engineering victim x Engineering negotiator	-0.684	0.561	0.223
Case configuration: Engineering victim x Control label hidden negotiator	0.312	0.545	0.567
Observer role (ref = victim)	0.912	0.327	0.005
Engineering participant (ref = humanities)	1.377	0.914	0.132
Man (ref = woman)	0.241	0.946	0.799
Age	0.368	0.195	0.059
Socioeconomic status	0.126	0.393	0.748
Negotiator 2 (ref = negotiator 1)	0.532	0.255	0.037
Scale variance (log)	1.876	0.057	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -12.864 (p=0.005). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β

coefficient of -0.930. The p-value of 0.247 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.523. The p-value of 0.025 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.738. The p-value of 0.042 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.291. The p-value of 0.181 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -1.030. The p-value of 0.095 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of -0.094. The p-value of 0.865 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 0.071. The p-value of 0.912 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.684. The p-value of 0.223 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 0.312. The p-value of 0.567 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.912. The p-value of 0.005 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.377. The p-value of 0.132 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.241. The p-value of 0.799 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.368. The p-value of 0.059 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.126. The p-value of 0.748 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale

parameter log is 1.876, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.963, p = 9.425e-22). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 9: H1 Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-15.603	5.656	0.006
Empathy: Fantasy scale	-1.028	1.718	0.550
Empathy: Empathic concern	-0.246	1.118	0.826
Empathy: Perspective taking	0.884	0.940	0.347
Empathy: Personal distress	-0.405	0.896	0.651
Outgroup perpetrator (ref = ingroup)	-0.171	1.967	0.931
Control label hidden (ref = ingroup)	0.297	1.733	0.864
Victim outgroup (ref = ingroup)	0.144	0.209	0.491
Observer role (ref = victim)	0.847	0.285	0.003
Engineering participant (ref = humanities)	0.239	1.080	0.825
Man (ref = woman)	0.334	0.496	0.500
Age	0.591	0.220	0.007
Socioeconomic status	-0.062	0.480	0.898
Negotiator 2 (ref = negotiator 1)	0.190	0.190	0.317
Fantasy x outgroup perpetrator	0.675	1.467	0.645
Empathic concern x outgroup perpetrator	-2.055	0.554	0.000
Perspective taking x outgroup perpetrator	-1.015	1.187	0.392
Personal distress x outgroup perpetrator	2.891	0.787	0.000
Fantasy x control label hidden	0.138	1.313	0.916
Empathic concern x control label hidden	-0.706	0.407	0.083
Perspective taking x control label hidden	-0.110	0.983	0.910
Personal distress x control label hidden	0.823	0.983	0.403

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.603 (p=0.006). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.028. The p-value of 0.550 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for

this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.246. The p-value of 0.826 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.884. The p-value of 0.347 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.405. The p-value of 0.651 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.171. The p-value of 0.931 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.297. The p-value of 0.864 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.144. The p-value of 0.491 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.847. The p-value of 0.003 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.239. The p-value of 0.825 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.334. The p-value of 0.500 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.591. The p-value of 0.007 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.062. The p-value of 0.898 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of 0.675. The p-value of 0.645 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_outgroup (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.055. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt:perp_outgroup (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -1.015. The p-value of 0.392 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_outgroup (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.891. The p-value of 0.000 indicates highly statistically

significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of 0.138. The p-value of 0.916 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of -0.706. The p-value of 0.083 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of -0.110. The p-value of 0.910 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.823. The p-value of 0.403 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 4 bootstrap refits converged successfully. The full-sample optimization status is converged after 93 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.2 H2a: Relational Betrayal Contrasts

This section tests whether same-faculty and cross-faculty betrayal cases differ when they are represented directly as explicit victim x negotiator scenarios.

9.2.1 Model A: Composite Empathy Control

Table 10: H2a Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.882	4.501	0.002
Case configuration: Humanities victim x Engineering negotiator	-1.165	0.622	0.061
Case configuration: Engineering victim x Humanities negotiator	0.229	0.645	0.723
Case configuration: Engineering victim x Engineering negotiator	-0.551	0.569	0.332
Empathy composite (average)	-0.735	1.037	0.478
Observer role (ref = victim)	1.196	0.381	0.002
Engineering participant (ref = humanities)	0.859	0.928	0.355
Man (ref = woman)	0.446	0.936	0.634
Age	0.432	0.187	0.021
Socioeconomic status	0.471	0.392	0.229
Negotiator 2 (ref = negotiator 1)	0.435	0.336	0.196
Scale variance (log)	1.893	0.063	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.882 (p=0.002). The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -1.165. The p-value of 0.061 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 0.229. The p-value of 0.723 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.551. The p-value of 0.332 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.735. The p-value of 0.478 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.196. The p-value of 0.002 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.859. The p-value of 0.355 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.446. The p-value of 0.634 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.432. The p-value of 0.021 indicates

statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.471. The p-value of 0.229 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.893, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.948, p = 1.326e-20). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 11: H2a Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-16.137	NA	1
Case configuration: Humanities victim x Engineering negotiator	-0.766	NA	1
Case configuration: Engineering victim x Humanities negotiator	-0.192	NA	1
Case configuration: Engineering victim x Engineering negotiator	-0.388	NA	1
Empathy composite (average)	-0.304	NA	1
Observer role (ref = victim)	0.980	NA	1
Engineering participant (ref = humanities)	-0.353	NA	1
Man (ref = woman)	0.525	NA	1
Age	0.597	NA	1
Socioeconomic status	0.170	NA	1
Negotiator 2 (ref = negotiator 1)	0.246	NA	1

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -16.137 (p=1.000). The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -0.766. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of -0.192. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.388. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.304. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively

as an observer rather than a victim) has an estimated β coefficient of 0.980. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.353. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.525. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.597. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.170. The p-value of 1.000 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 1 ids with replacement, retaining all repeated observations from each sampled participant; 1 bootstrap refits converged successfully. The full-sample optimization status is converged after 46 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.2.2 Model B: Separated Construct Controls

Table 12: H2a Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.161	4.847	0.007
Case configuration: Humanities victim x Engineering negotiator	-1.026	0.619	0.097
Case configuration: Engineering victim x Humanities negotiator	0.271	0.645	0.674
Case configuration: Engineering victim x Engineering negotiator	-0.494	0.565	0.382
Empathy: Fantasy scale	-0.952	0.860	0.268
Empathy: Empathic concern	-2.825	1.169	0.016
Empathy: Perspective taking	1.627	0.968	0.093
Empathy: Personal distress	1.582	0.968	0.102
Observer role (ref = victim)	1.133	0.378	0.003
Engineering participant (ref = humanities)	0.851	0.952	0.371
Man (ref = woman)	0.265	0.989	0.789
Age	0.393	0.194	0.042
Socioeconomic status	0.320	0.407	0.431
Negotiator 2 (ref = negotiator 1)	0.265	0.328	0.419
Scale variance (log)	1.876	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.161 (p=0.007). The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -1.026. The p-value of 0.097 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 0.271. The p-value of 0.674 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.494. The p-value of 0.382 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.952. The p-value of 0.268 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.825. The p-value of 0.016 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.627. The p-value of 0.093 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.582. The p-value of 0.102 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.133. The p-value of 0.003 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.851. The p-value of 0.371 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.265. The p-value of 0.789 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.393. The p-value of 0.042 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.320. The p-value of 0.431 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.876, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption

(W = 0.961, p = 6.381e-18). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 13: H2a Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-16.182	6.376	0.011
Negotiator and victim share faculty	0.133	0.188	0.477
Empathy: Fantasy scale	-0.599	1.056	0.570
Empathy: Empathic concern	-1.178	0.893	0.187
Empathy: Perspective taking	0.245	0.805	0.760
Empathy: Personal distress	0.852	0.637	0.181
Outgroup perpetrator (ref = ingroup)	-0.114	0.491	0.816
Victim outgroup (ref = ingroup)	0.134	0.312	0.668
Observer role (ref = victim)	1.113	0.498	0.025
Engineering participant (ref = humanities)	-0.039	0.985	0.969
Man (ref = woman)	0.439	0.506	0.386
Age	0.605	0.261	0.020
Socioeconomic status	0.236	0.328	0.471
Negotiator 2 (ref = negotiator 1)	0.258	0.454	0.570

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -16.182 (p=0.011). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of 0.133. The p-value of 0.477 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.599. The p-value of 0.570 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.178. The p-value of 0.187 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.245. The p-value of 0.760 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.852. The p-value of 0.181 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for

this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.114. The p-value of 0.816 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.134. The p-value of 0.668 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.113. The p-value of 0.025 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.039. The p-value of 0.969 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.439. The p-value of 0.386 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.605. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.236. The p-value of 0.471 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 4 bootstrap refits converged successfully. The full-sample optimization status is converged after 27 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.3 H2b: Explicit Case-Configuration Contrasts

This section examines interpretable victim x negotiator case contrasts directly, replacing isolated outgroup-perpetrator indicators with explicit relational scenarios.

9.3.1 Model A: Composite Empathy Control

Table 14: H2b Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.201	4.407	0.003
Case configuration: Humanities victim x Engineering negotiator	-1.145	0.621	0.065
Case configuration: Humanities victim x Control label hidden negotiator	-0.050	0.551	0.928
Case configuration: Engineering victim x Humanities negotiator	0.013	0.641	0.983
Case configuration: Engineering victim x Engineering negotiator	-0.757	0.564	0.179
Case configuration: Engineering victim x Control label hidden negotiator	0.317	0.554	0.566
Empathy composite (average)	-0.671	1.002	0.503
Observer role (ref = victim)	0.929	0.324	0.004
Engineering participant (ref = humanities)	1.415	0.889	0.111
Man (ref = woman)	0.356	0.892	0.690
Age	0.408	0.193	0.035
Socioeconomic status	0.267	0.392	0.495
Negotiator 2 (ref = negotiator 1)	0.621	0.261	0.017
Scale variance (log)	1.891	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.201 (p=0.003). The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -1.145. The p-value of 0.065 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of -0.050. The p-value of 0.928 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 0.013. The p-value of 0.983 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.757. The p-value of 0.179 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 0.317. The p-value of 0.566 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.671. The p-value of 0.503 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.929. The p-value of 0.004 indicates statistically significant, meaning the predictor

has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.415. The p-value of 0.111 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.356. The p-value of 0.690 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.408. The p-value of 0.035 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.267. The p-value of 0.495 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.891, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.954, p = 3.779e-24). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 15: H2b Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-15.801	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator	-0.892	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator	-0.164	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator	-0.147	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator	-0.418	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator	0.090	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	-0.447	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.999	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.129	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.175	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.615	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.012	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.271	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.801 (p=NA). The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of -0.892. The p-value of NA the p-value cannot be determined. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of -0.164. The p-value of NA the p-value cannot be determined. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of -0.147. The p-value of NA the p-value cannot be determined. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.418. The p-value of NA the p-value cannot be determined. The term case_ing_x_control (rep-

resenting whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 0.090. The p-value of NA the p-value cannot be determined. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.447. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.999. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.129. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.175. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.615. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.012. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.3.2 Model B: Separated Construct Controls

Table 16: H2b Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-12.864	4.619	0.005
Case configuration: Humanities victim x Engineering negotiator	-1.030	0.617	0.095
Case configuration: Humanities victim x Control label hidden negotiator	-0.094	0.552	0.865
Case configuration: Engineering victim x Humanities negotiator	0.071	0.639	0.912
Case configuration: Engineering victim x Engineering negotiator	-0.684	0.561	0.223
Case configuration: Engineering victim x Control label hidden negotiator	0.312	0.545	0.567
Empathy: Fantasy scale	-0.930	0.804	0.247
Empathy: Empathic concern	-2.523	1.123	0.025
Empathy: Perspective taking	1.738	0.857	0.042
Empathy: Personal distress	1.291	0.965	0.181
Observer role (ref = victim)	0.912	0.327	0.005
Engineering participant (ref = humanities)	1.377	0.914	0.132
Man (ref = woman)	0.241	0.946	0.799
Age	0.368	0.195	0.059
Socioeconomic status	0.126	0.393	0.748
Negotiator 2 (ref = negotiator 1)	0.532	0.255	0.037
Scale variance (log)	1.876	0.057	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -12.864 (p=0.005). The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has

an estimated β coefficient of -1.030. The p-value of 0.095 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of -0.094. The p-value of 0.865 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 0.071. The p-value of 0.912 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.684. The p-value of 0.223 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 0.312. The p-value of 0.567 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.930. The p-value of 0.247 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.523. The p-value of 0.025 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.738. The p-value of 0.042 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.291. The p-value of 0.181 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.912. The p-value of 0.005 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.377. The p-value of 0.132 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.241. The p-value of 0.799 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.368. The p-value of 0.059 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.126. The p-value of 0.748 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale

parameter log is 1.876, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.963, p = 9.425e-22). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 17: H2b Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-15.603	5.169	0.003
Outgroup perpetrator (ref = ingroup)	-0.171	2.676	0.949
Empathy: Fantasy scale	-1.028	1.044	0.325
Empathy: Empathic concern	-0.246	0.575	0.668
Empathy: Perspective taking	0.884	0.828	0.286
Empathy: Personal distress	-0.405	0.925	0.662
Control label hidden (ref = ingroup)	0.297	1.792	0.869
Victim outgroup (ref = ingroup)	0.144	0.370	0.698
Observer role (ref = victim)	0.847	0.230	0.000
Engineering participant (ref = humanities)	0.239	1.756	0.892
Man (ref = woman)	0.334	0.814	0.681
Age	0.591	0.244	0.015
Socioeconomic status	-0.062	0.358	0.863
Negotiator 2 (ref = negotiator 1)	0.190	0.137	0.164
Fantasy x outgroup perpetrator	0.675	0.867	0.436
Empathic concern x outgroup perpetrator	-2.055	0.420	0.000
Perspective taking x outgroup perpetrator	-1.015	0.284	0.000
Personal distress x outgroup perpetrator	2.891	0.650	0.000
Fantasy x control label hidden	0.138	0.444	0.757
Empathic concern x control label hidden	-0.706	0.434	0.104
Perspective taking x control label hidden	-0.110	0.541	0.838
Personal distress x control label hidden	0.823	0.748	0.272

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.603 (p=0.003). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.171. The p-value of 0.949 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of

fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.028. The p-value of 0.325 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.246. The p-value of 0.668 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.884. The p-value of 0.286 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.405. The p-value of 0.662 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.297. The p-value of 0.869 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.144. The p-value of 0.698 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.847. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.239. The p-value of 0.892 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.334. The p-value of 0.681 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.591. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.062. The p-value of 0.863 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_fs (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of 0.675. The p-value of 0.436 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.055. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -1.015. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.891. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the

combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of 0.138. The p-value of 0.757 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of -0.706. The p-value of 0.104 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of -0.110. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.823. The p-value of 0.272 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 3 bootstrap refits converged successfully. The full-sample optimization status is converged after 93 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.4 H3: Interaction Moderation

This section tests whether empathy is conditioned by explicit case-configuration contrasts rather than by a single outgroup flag.

9.4.1 Model A: Composite Empathy Interaction

Table 18: H3 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-16.704	4.841	0.001
Empathy composite (average)	0.879	1.364	0.519
Case configuration: Humanities victim x Engineering negotiator	5.011	3.307	0.130
Case configuration: Humanities victim x Control label hidden negotiator	1.542	2.564	0.548
Case configuration: Engineering victim x Humanities negotiator	5.718	3.344	0.087
Case configuration: Engineering victim x Engineering negotiator	1.800	3.141	0.567
Case configuration: Engineering victim x Control label hidden negotiator	4.807	3.315	0.147
Observer role (ref = victim)	0.943	0.330	0.004
Engineering participant (ref = humanities)	1.431	0.880	0.104
Man (ref = woman)	0.334	0.884	0.705
Age	0.409	0.191	0.032
Socioeconomic status	0.268	0.391	0.493
Negotiator 2 (ref = negotiator 1)	0.589	0.262	0.025
Case configuration: Humanities victim x Engineering negotiator x Empathy composite (average)	-2.738	1.390	0.049
Case configuration: Humanities victim x Control label hidden negotiator x Empathy composite (average)	-0.693	1.114	0.534
Case configuration: Engineering victim x Humanities negotiator x Empathy composite (average)	-2.532	1.444	0.080
Case configuration: Engineering victim x Engineering negotiator x Empathy composite (average)	-1.140	1.348	0.398
Case configuration: Engineering victim x Control label hidden negotiator x Empathy composite (average)	-1.994	1.450	0.169
Scale variance (log)	1.889	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -16.704 (p=0.001). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.879. The p-value of 0.519 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of 5.011. The p-value of 0.130 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of 1.542. The p-value of 0.548 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 5.718. The p-value of 0.087 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of 1.800. The p-value of 0.567 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 4.807. The p-value of 0.147 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.943. The p-value of 0.004 indicates statistically significant, meaning the predictor

has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.431. The p-value of 0.104 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.334. The p-value of 0.705 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.409. The p-value of 0.032 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.268. The p-value of 0.493 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -2.738. The p-value of 0.049 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_total:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.693. The p-value of 0.534 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -2.532. The p-value of 0.080 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.140. The p-value of 0.398 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.994. The p-value of 0.169 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.889, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.955, p = 4.559e-24). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 19: H3 Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-18.758	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	1.010	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator	2.749	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator	2.984	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator	4.249	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator	1.801	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator	3.649	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.999	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.143	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.186	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.602	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.020	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.343	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator x Empathy composite (average)	-1.712	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator x Empathy composite (average)	-1.496	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator x Empathy composite (average)	-2.056	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator x Empathy composite (average)	-1.055	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator x Empathy composite (average)	-1.614	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -18.758 (p=NA). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 1.010. The p-value of NA the p-value cannot be determined. The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of 2.749. The p-value of NA the p-value cannot be determined. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of 2.984. The p-value of NA the p-value cannot be determined. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 4.249. The p-value of NA the p-value cannot be determined. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of 1.801. The p-value of NA the p-value cannot be determined. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 3.649. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.999. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.143. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.186. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.602. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.020. The p-value of NA the p-value cannot be determined. The term iri_total:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.712. The p-value of NA the p-value cannot be determined. The term iri_total:case_hum_x_control (representing whether the judged scenario matches the

explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.496. The p-value of NA the p-value cannot be determined. The term iri_total:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -2.056. The p-value of NA the p-value cannot be determined. The term iri_total:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.055. The p-value of NA the p-value cannot be determined. The term iri_total:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.614. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.4.2 Model B: Separated Constructs Interaction

Table 20: H3 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-15.990	5.251	0.002
Empathy: Fantasy scale	-0.552	0.955	0.563
Empathy: Empathic concern	-0.294	1.365	0.830
Empathy: Perspective taking	1.151	1.024	0.261
Empathy: Personal distress	0.753	1.165	0.518
Case configuration: Humanities victim x Engineering negotiator	5.420	3.836	0.158
Case configuration: Humanities victim x Control label hidden negotiator	0.120	2.836	0.966
Case configuration: Engineering victim x Humanities negotiator	5.717	4.048	0.158
Case configuration: Engineering victim x Engineering negotiator	1.088	3.373	0.747
Case configuration: Engineering victim x Control label hidden negotiator	4.526	3.670	0.218
Observer role (ref = victim)	0.956	0.347	0.006
Engineering participant (ref = humanities)	1.388	0.892	0.120
Man (ref = woman)	0.156	0.930	0.867
Age	0.363	0.194	0.061
Socioeconomic status	0.128	0.392	0.744
Negotiator 2 (ref = negotiator 1)	0.484	0.254	0.057
Case configuration: Humanities victim x Engineering negotiator x Empathy: Fantasy scale	-0.525	1.115	0.637
Case configuration: Humanities victim x Engineering negotiator x Empathy: Empathic concern	-1.660	1.737	0.339
Case configuration: Humanities victim x Engineering negotiator x Empathy: Perspective taking	-0.587	1.217	0.630
Case configuration: Humanities victim x Engineering negotiator x Empathy: Personal distress	-0.051	1.155	0.964
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Fantasy scale	-1.105	0.916	0.228
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Empathic concern	0.820	1.514	0.588
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Perspective taking	0.610	0.923	0.509
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Personal distress	-0.680	1.248	0.586
Case configuration: Engineering victim x Humanities negotiator x Empathy: Fantasy scale	-0.778	1.374	0.571
Case configuration: Engineering victim x Humanities negotiator x Empathy: Empathic concern	-5.682	1.574	0.000
Case configuration: Engineering victim x Humanities negotiator x Empathy: Perspective taking	0.833	1.674	0.619
Case configuration: Engineering victim x Humanities negotiator x Empathy: Personal distress	3.318	1.368	0.015
Case configuration: Engineering victim x Engineering negotiator x Empathy: Fantasy scale	0.052	1.220	0.966
Case configuration: Engineering victim x Engineering negotiator x Empathy: Empathic concern	-1.375	1.321	0.298
Case configuration: Engineering victim x Engineering negotiator x Empathy: Perspective taking	1.207	1.258	0.337
Case configuration: Engineering victim x Engineering negotiator x Empathy: Personal distress	-0.845	1.246	0.498
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Fantasy scale	-0.105	1.229	0.932
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Empathic concern	-3.135	1.633	0.055
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Perspective taking	0.780	1.447	0.590
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Personal distress	0.605	1.406	0.667
Scale variance (log)	1.863	0.057	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -15.990 (p=0.002). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.552. The p-value of 0.563 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.294. The p-value of 0.830 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.151. The p-value of 0.261 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.753. The p-value

of 0.518 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of 5.420. The p-value of 0.158 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of 0.120. The p-value of 0.966 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 5.717. The p-value of 0.158 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of 1.088. The p-value of 0.747 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 4.526. The p-value of 0.218 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.956. The p-value of 0.006 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.388. The p-value of 0.120 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.156. The p-value of 0.867 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.363. The p-value of 0.061 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.128. The p-value of 0.744 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.525. The p-value of 0.637 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.660. The p-value of 0.339 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:case_hum_x_ing (representing whether the judged scenario matches the explicit

victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of -0.587. The p-value of 0.630 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.051. The p-value of 0.964 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.105. The p-value of 0.228 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of 0.820. The p-value of 0.588 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.610. The p-value of 0.509 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.680. The p-value of 0.586 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.778. The p-value of 0.571 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -5.682. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.833. The p-value of 0.619 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x

Humanities negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 3.318. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of 0.052. The p-value of 0.966 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.375. The p-value of 0.298 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.207. The p-value of 0.337 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.845. The p-value of 0.498 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.105. The p-value of 0.932 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -3.135. The p-value of 0.055 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.780. The p-value of 0.590 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.605. The p-value of 0.667 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.863, capturing the underlying dispersion of latent

judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.967, p = 9.484e-21). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 21: H3 Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-16.454	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Fantasy scale	-0.262	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Empathic concern	0.348	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Perspective taking	-0.539	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Personal distress	0.905	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator	1.400	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator	0.761	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator	4.894	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator	-0.345	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator	1.943	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.964	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	0.110	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.107	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.568	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	-0.073	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.327	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator x Empathy: Fantasy scale	-1.016	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator x Empathy: Empathic concern	-1.386	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator x Empathy: Perspective taking	0.283	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Engineering negotiator x Empathy: Personal distress	0.838	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Fantasy scale	-1.668	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Empathic concern	0.236	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Perspective taking	1.494	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Humanities victim x Control label hidden negotiator x Empathy: Personal distress	-1.119	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator x Empathy: Fantasy scale	-0.449	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator x Empathy: Empathic concern	-4.728	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator x Empathy: Perspective taking	1.102	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Humanities negotiator x Empathy: Personal distress	2.111	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator x Empathy: Fantasy scale	-0.684	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator x Empathy: Empathic concern	-0.416	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator x Empathy: Perspective taking	2.079	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Engineering negotiator x Empathy: Personal distress	-1.572	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Fantasy scale	-0.486	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Empathic concern	-1.924	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Perspective taking	1.615	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Case configuration: Engineering victim x Control label hidden negotiator x Empathy: Personal distress	-0.324	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -16.454 (p=NA). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.262. The p-value of NA the p-value cannot be determined. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of 0.348. The p-value of NA the p-value cannot be determined. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of -0.539. The p-value of NA the p-value cannot be determined. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.905. The p-value of NA the p-value cannot be determined. The term case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator) has an estimated β coefficient of 1.400. The p-value of NA the p-value cannot be determined. The term case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator) has an estimated β coefficient of 0.761. The p-value of NA the p-value cannot be

determined. The term case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator) has an estimated β coefficient of 4.894. The p-value of NA the p-value cannot be determined. The term case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator) has an estimated β coefficient of -0.345. The p-value of NA the p-value cannot be determined. The term case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator) has an estimated β coefficient of 1.943. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.964. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.110. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.107. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.568. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.073. The p-value of NA the p-value cannot be determined. The term iri_fs:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.016. The p-value of NA the p-value cannot be determined. The term iri_ec:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.386. The p-value of NA the p-value cannot be determined. The term iri_pt:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.283. The p-value of NA the p-value cannot be determined. The term iri_pd:case_hum_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Engineering negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.838. The p-value of NA the p-value cannot be determined. The term iri_fs:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.668. The p-value of NA the p-value cannot be determined. The term iri_ec:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of 0.236. The p-value of NA the p-value cannot be determined. The term iri_pt:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.494.

The p-value of NA the p-value cannot be determined. The term iri_pd:case_hum_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Humanities victim x Control label hidden negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.119. The p-value of NA the p-value cannot be determined. The term iri_fs:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.449. The p-value of NA the p-value cannot be determined. The term iri_ec:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -4.728. The p-value of NA the p-value cannot be determined. The term iri_pt:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.102. The p-value of NA the p-value cannot be determined. The term iri_pd:case_ing_x_hum (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Humanities negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 2.111. The p-value of NA the p-value cannot be determined. The term iri_fs:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.684. The p-value of NA the p-value cannot be determined. The term iri_ec:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.416. The p-value of NA the p-value cannot be determined. The term iri_pt:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.079. The p-value of NA the p-value cannot be determined. The term iri_pd:case_ing_x_ing (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Engineering negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.572. The p-value of NA the p-value cannot be determined. The term iri_fs:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.486. The p-value of NA the p-value cannot be determined. The term iri_ec:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.924. The p-value of NA the p-value cannot be determined. The term iri_pt:case_ing_x_control (representing whether the judged scenario

matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.615. The p-value of NA the p-value cannot be determined. The term iri_pd:case_ing_x_control (representing whether the judged scenario matches the explicit victim x negotiator case configuration Engineering victim x Control label hidden negotiator interacted with the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.324. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

10 Discussion and Limitations

While the dual-estimator strategy strengthens the analysis, several limitations remain. First, both inference strategies assume independence between participants, which may still be violated by broader shared institutional or contextual shocks. Second, we assume a normal distribution for the latent errors ϵ_{ij} in the Tobit branch; departures from normality could affect the consistency of the maximum likelihood estimators. The non-parametric robustness branch reduces dependence on that assumption and adds participant-level cluster bootstrap inference after converged full-sample fits, but bootstrap summaries still carry Monte Carlo error and can be sensitive to convergence problems in difficult specifications. Finally, the use of unstandardized averages for empathy assumes a linear mapping between the psychometric scale and the latent moral preference across both estimators.

11 Conclusion

Based on the combined interval-censored Tobit estimations and the cluster-aware non-parametric robustness workflow, empathy and relational victim x negotiator case configurations have been documented together under Option 2 explicit case-configuration modeling, alongside their theoretical assumptions above.